

GRADE 4 COMMON CORE MATH

Answer Key

Main • Backward Branches • Forward 1 (Grade 5) • Forward 2 (Grade 6)

64 question sets • 1600 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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Interpret a multiplication equation as a comparison

MAIN	Grade 4 main question	4.OA.A.1
BACKWARD 1	Products as equal groups	3.OA.A.1
BACKWARD 2	Multiplication facts within 100	3.OA.C.7
FORWARD 1	Compare the size of a product to the size of one factor	5.NF.B.5.a
FORWARD 2	Use ratio language to describe a relationship	6.RP.A.1

SET 1 • 4.OA.A.1

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1

Which statement matches $24 = 6 \times 4$?

- A.** 24 is 6 more than 4.
- B.** 6 is 4 times as many as 24.
- C.** 24 is 4 more than 6.
- D.** 24 is 6 times as many as 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • MAIN • Q1/5

8

QUESTION ID

S1-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. 24 is 6 times as many as 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • MAIN • Q1/5

Write a comparison statement for $45 = 9 \times 5$.

Statement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • MAIN • Q2/5

9

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

45 is 9 times as many as 5.

GRADING NOTE "45 is 5 times as many as 9" is also correct. A statement with "more than" is not correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • MAIN • Q2/5

True or false?

$56 = 7 \times 8$ can mean that 56 is 8 times as many as 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • MAIN • Q3/5

10

QUESTION ID

S1-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • MAIN • Q3/5

True or false?

$18 = 3 \times 6$ means that 18 is 3 more than 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • MAIN • Q4/5

11

QUESTION ID

S1-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • MAIN • Q4/5

Which equation means "63 is 7 times as many as 9"?

A. $63 = 7 + 9$

B. $7 = 63 \times 9$

C. $63 = 7 \times 9$

D. $63 = 9 - 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • MAIN • Q5/5

12

QUESTION ID

S1-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. $63 = 7 \times 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 4.OA.A.1

BACKWARD 1

Products as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

What is the total in 4 groups of 6?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • BACKWARD 1 • Q1/5

14

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • BACKWARD 1 • Q1/5

True or false?

5×3 can mean 5 groups with 3 objects in each group.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • BACKWARD 1 • Q2/5

15

QUESTION ID

S1-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • BACKWARD 1 • Q2/5

Which expression shows 3 groups of 7?

- A. $7 - 3$
- B. $3 + 7$
- C. $7 \div 3$
- D. 3×7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • BACKWARD 1 • Q3/5

16

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3×7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • BACKWARD 1 • Q3/5

Write a multiplication equation for 6 groups of 2.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • BACKWARD 1 • Q4/5

17

QUESTION ID

S1-B1-Q4

TYPE

Short answer

CORRECT ANSWER

6 × 2 = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • BACKWARD 1 • Q4/5

True or false?

2×8 means 2 more than 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • BACKWARD 1 • Q5/5

18

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 4.OA.A.1

BACKWARD 2

Multiplication facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$9 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q1/5

20

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q1/5

True or false?

$$7 \times 8 = 54$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q2/5

21

QUESTION ID

S1-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q2/5

$$6 \times 4 = ?$$

- A.** 10
- B.** 20
- C.** 24
- D.** 28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q3/5

22

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q3/5

Multiply.

$$8 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q4/5

23

QUESTION ID

S1-B2-Q4

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q4/5

Multiply.

$$7 \times 9$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q5/5

24

QUESTION ID

S1-B2-Q5

TYPE

Short answer

CORRECT ANSWER

63

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 4.OA.A.1

FORWARD 1

Compare the size of a product to the size of one factor • Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a

Without multiplying, how does 3×25 compare with 25?

- A. It is 3 more.
- B. It is smaller than 25.
- C. It is equal to 25.
- D. It is 3 times as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 1 • FORWARD 1 • Q1/5

26

QUESTION ID

S1-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. It is 3 times as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 1 • FORWARD 1 • Q1/5

True or false?

$\frac{1}{2} \times 40$ is less than 40.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 1 • FORWARD 1 • Q2/5

27

QUESTION ID

S1-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 1 • FORWARD 1 • Q2/5

A rope is 12 feet long. A second rope is 4 times as long.
How long is the second rope?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 1 • FORWARD 1 • Q3/5

28

QUESTION ID

S1-F1-Q3

TYPE

Short answer

CORRECT ANSWER

48 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 1 • FORWARD 1 • Q3/5

True or false?

5×18 is less than 18.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 1 • FORWARD 1 • Q4/5

29

QUESTION ID

S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 1 • FORWARD 1 • Q4/5

Which product is greater than 30?

A. $\frac{1}{2} \times 30$

B. 2×30

C. $\frac{3}{4} \times 30$

D. $\frac{1}{3} \times 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 1 • FORWARD 1 • Q5/5

30

QUESTION ID

S1-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 2×30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 1 • FORWARD 1 • Q5/5

SET 1 • 4.OA.A.1

FORWARD 2

Use ratio language to describe a relationship • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

A class has 12 girls and 4 boys.

Write the ratio of girls to boys.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • FORWARD 2 • Q1/5

32

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

12 : 4

GRADING NOTE 3 : 1, 12 to 4, and 12/4 are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • FORWARD 2 • Q1/5

True or false?

If there are 3 times as many cats as dogs, the ratio of cats to dogs is 3 : 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • FORWARD 2 • Q2/5

33

QUESTION ID

S1-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • FORWARD 2 • Q2/5

A bag has 5 red marbles for every 2 blue marbles.

Which statement describes the ratio?

- A. There are 5 more red marbles than blue.
- B. For every 5 red marbles there are 2 blue marbles.
- C. There are 2 times as many red marbles as blue.
- D. There are 7 red marbles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • FORWARD 2 • Q3/5

34

QUESTION ID

S1-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. For every 5 red marbles there are 2 blue marbles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • FORWARD 2 • Q3/5

There are 18 apples and 6 pears.

How many apples are there for every 1 pear?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • FORWARD 2 • Q4/5

35

QUESTION ID

S1-F2-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

The ratio 2 : 8 means there are 2 times as many of the first quantity as the second.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • FORWARD 2 • Q5/5

36

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • FORWARD 2 • Q5/5

Write a comparison statement as a multiplication equation

MAIN	Grade 4 main question	4.OA.A.1
BACKWARD 1	Multiplication word problems within 100	3.OA.A.3
BACKWARD 2	Find the unknown number in a multiplication equation	3.OA.A.4
FORWARD 1	Write numerical expressions from words	5.OA.A.2
FORWARD 2	Write expressions that use a variable	6.EE.A.2.a

SET 2 • 4.OA.A.1

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1

Write an equation.

32 is 4 times as many as 8.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • MAIN • Q1/5

39

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

$32 = 4 \times 8$

GRADING NOTE $32 = 8 \times 4$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • MAIN • Q1/5

Kai has 5 stickers. Lena has 3 times as many stickers as Kai.

Which equation gives Lena's stickers, s ?

A. $s = 3 \times 5$

B. $s = 5 - 3$

C. $s = 3 + 5$

D. $s = 5 \div 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • MAIN • Q2/5

40

QUESTION ID

S2-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $s = 3 \times 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • MAIN • Q2/5

Write an equation.

54 is 6 times as many as 9.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • MAIN • Q3/5

41

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

54 = 6 × 9

GRADING NOTE $54 = 9 \times 6$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • MAIN • Q3/5

True or false?

"40 is 8 times as many as 5" can be written as
 $40 = 8 + 5$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • MAIN • Q4/5

42

QUESTION ID

S2-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • MAIN • Q4/5

A tree is 7 times as tall as a 2-meter bush.

Write an equation for the tree's height, t .

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • MAIN • Q5/5

43

QUESTION ID

S2-M-Q5

TYPE

Short answer

CORRECT ANSWER

$$t = 7 \times 2$$

GRADING NOTE $t = 2 \times 7$ is also correct. (The tree is 14 meters tall.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • MAIN • Q5/5

SET 2 • 4.OA.A.1

BACKWARD 1

Multiplication word problems within 100 • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

There are 6 boxes with 8 crayons in each box.

How many crayons are there?

Crayons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 2 • BACKWARD 1 • Q1/5

45

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 2 • BACKWARD 1 • Q1/5

True or false?

5 bags with 9 apples in each bag hold 45 apples.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 2 • BACKWARD 1 • Q2/5

46

QUESTION ID

S2-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 2 • BACKWARD 1 • Q2/5

A garden has 4 rows with 7 plants in each row.

How many plants are there?

- A.** 21
- B.** 32
- C.** 28
- D.** 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 2 • BACKWARD 1 • Q3/5

47

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 2 • BACKWARD 1 • Q3/5

Each pack has 3 juice boxes.

How many juice boxes are in 9 packs?

Juice boxes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 2 • BACKWARD 1 • Q4/5

48

QUESTION ID

S2-B1-Q4

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 2 • BACKWARD 1 • Q4/5

True or false?

8 teams with 5 players each have 13 players in all.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 2 • BACKWARD 1 • Q5/5

49

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 4.OA.A.1

BACKWARD 2

Find the unknown number in a multiplication equation • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

$$? \times 6 = 30$$

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 2 • BACKWARD 2 • Q1/5

51

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 2 • BACKWARD 2 • Q1/5

True or false?

If $4 \times n = 36$, then $n = 8$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 2 • BACKWARD 2 • Q2/5

52

QUESTION ID

S2-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 2 • BACKWARD 2 • Q2/5

$$8 \times ? = 56$$

- A.** 8
- B.** 7
- C.** 6
- D.** 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 2 • BACKWARD 2 • Q3/5

53

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 2 • BACKWARD 2 • Q3/5

Find the unknown number.

$$3 \times ? = 27$$

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 2 • BACKWARD 2 • Q4/5

54

QUESTION ID

S2-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 2 • BACKWARD 2 • Q4/5

True or false?

If $m \times 7 = 42$, then $m = 6$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 2 • BACKWARD 2 • Q5/5

55

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 4.OA.A.1

FORWARD 1

Write numerical expressions from words • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression.

3 times the sum of 8 and 4

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 1 • Q1/5

57

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$3 \times (8 + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 1 • Q1/5

Which expression means "add 9 and 6, then multiply by 2"?

- A. $9 \times 6 + 2$
- B. $9 + 6 \times 2$
- C. $(9 + 6) \times 2$
- D. $2 + 9 \times 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 1 • Q2/5

58

QUESTION ID

S2-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $(9 + 6) \times 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 1 • Q2/5

True or false?

$5 \times (12 - 3)$ is 5 times as large as $12 - 3$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 1 • Q3/5

59

QUESTION ID

S2-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 1 • Q3/5

Write an expression.

4 times as large as $25 + 15$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 1 • Q4/5

60

QUESTION ID

S2-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$4 \times (25 + 15)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 1 • Q4/5

True or false?

$2 \times (7 + 1)$ is 2 more than $7 + 1$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 1 • Q5/5

61

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 4.OA.A.1

FORWARD 2

Write expressions that use a variable • Grade 6 • Expressions & Equations •
6.EE.A.2.a

Write an expression for 5 times a number n .

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 2 • FORWARD 2 • Q1/5

63

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5n

GRADING NOTE $5 \times n$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 2 • FORWARD 2 • Q1/5

Which expression means "8 more than y "?

- A. $y + 8$
- B. $8 - y$
- C. $y - 8$
- D. $8y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 2 • FORWARD 2 • Q2/5

64

QUESTION ID

S2-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $y + 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 2 • FORWARD 2 • Q2/5

Maya has m marbles. Jo has 4 times as many marbles as Maya.

Write an expression for Jo's marbles.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 2 • FORWARD 2 • Q3/5

65

QUESTION ID

S2-F2-Q3

TYPE

Short answer

CORRECT ANSWER

4m

GRADING NOTE $4 \times m$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 2 • FORWARD 2 • Q3/5

True or false?

"3 times the sum of x and 2" can be written as $3(x + 2)$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 2 • FORWARD 2 • Q4/5

66

QUESTION ID

S2-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 2 • FORWARD 2 • Q4/5

True or false?

"7 less than k" can be written as $7 - k$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 2 • FORWARD 2 • Q5/5

67

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 2 • FORWARD 2 • Q5/5

Solve multiplicative comparison word problems by multiplying

MAIN	Grade 4 main question	4.OA.A.2
BACKWARD 1	Multiplication and division word problems within 100	3.OA.A.3
BACKWARD 2	Multiply in either order	3.OA.B.5
FORWARD 1	Solve real-world problems by multiplying by a fraction or mixed nu...	5.NF.B.6
FORWARD 2	Solve unit rate problems	6.RP.A.3.b

SET 3 • 4.OA.A.2

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A puppy weighs 6 pounds. Its mother weighs 8 times as much.

How much does the mother weigh?

Puppy

6

Mother

6

6

6

6

6

6

6

6

Weight: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • MAIN • Q1/5

70

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

48 pounds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • MAIN • Q1/5

Sam read 9 pages. Ana read 5 times as many pages as Sam.

How many pages did Ana read?

Pages: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • MAIN • Q2/5

71

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • MAIN • Q2/5

A red ribbon is 7 inches long. A blue ribbon is 6 times as long.

How long is the blue ribbon?

- A.** 13 inches **B.** 48 inches
C. 1 inch **D.** 42 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • MAIN • Q3/5

72

QUESTION ID

S3-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 42 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • MAIN • Q3/5

A small tank holds 9 gallons. A large tank holds 7 times as much.

How much does the large tank hold?

Gallons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • MAIN • Q4/5

73

QUESTION ID

S3-M-Q4

TYPE

Short answer

CORRECT ANSWER

63

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • MAIN • Q4/5

True or false?

A bike costs \$80. A used car costs 100 times as much. The car costs \$800.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • MAIN • Q5/5

74

QUESTION ID

S3-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • MAIN • Q5/5

SET 3 • 4.OA.A.2

BACKWARD 1

Multiplication and division word problems within 100 • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

7 vans each carry 8 students.

How many students are there in all?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 3 • BACKWARD 1 • Q1/5

76

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 3 • BACKWARD 1 • Q1/5

36 crayons are shared equally by 4 children.

How many crayons does each child get?

Crayons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 3 • BACKWARD 1 • Q2/5

77

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 3 • BACKWARD 1 • Q2/5

True or false?

6 rows of 6 chairs make 36 chairs.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 3 • BACKWARD 1 • Q3/5

78

QUESTION ID

S3-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 3 • BACKWARD 1 • Q3/5

A book has 9 chapters with 9 pages in each chapter.

How many pages does it have?

- A. 18
- B. 72
- C. 81
- D. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 3 • BACKWARD 1 • Q4/5

79

QUESTION ID

S3-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 3 • BACKWARD 1 • Q4/5

True or false?

40 cards in 5 equal piles makes 9 cards in each pile.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 3 • BACKWARD 1 • Q5/5

80

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 4.OA.A.2

BACKWARD 2

Multiply in either order • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$6 \times 8 = 8 \times 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q1/5

82

QUESTION ID

S3-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q1/5

Fill in the blank.

$$9 \times 4 = 4 \times \underline{\quad}$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q2/5

83

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q2/5

Which is equal to 7×5 ?

- A.** 5×7
- B.** $7 + 5$
- C.** 75
- D.** $5 + 5 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q3/5

84

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5×7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

$$3 \times 9 = 27.$$

What is 9×3 ?

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q4/5

85

QUESTION ID

S3-B2-Q4

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q4/5

85

True or false?

$$8 \times 2 = 2 + 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q5/5

86

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • 4.OA.A.2

FORWARD 1

Solve real-world problems by multiplying by a fraction or mixed number •
Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A pup weighs 6 pounds. An adult dog weighs $4\frac{1}{2}$ times as much.

How much does the adult weigh?

Weight: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 3 • FORWARD 1 • Q1/5

88

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

27 pounds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 3 • FORWARD 1 • Q1/5

A path is 12 miles long. Kim walked $\frac{2}{3}$ of the path.

How far did she walk?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 3 • FORWARD 1 • Q2/5

89

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

8 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 3 • FORWARD 1 • Q2/5

A red rope is 8 feet long. A blue rope is $1\frac{1}{4}$ times as long.

How long is the blue rope?

- A.** 10 feet **B.** $9\frac{1}{4}$ feet
C. 2 feet **D.** 6 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 3 • FORWARD 1 • Q3/5

90

QUESTION ID

S3-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 3 • FORWARD 1 • Q3/5

True or false?

A 10-cm plant grows until it is $2\frac{1}{2}$ times as tall. It is then 25 cm tall.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 3 • FORWARD 1 • Q4/5

91

QUESTION ID

S3-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 3 • FORWARD 1 • Q4/5

A recipe uses 3 cups of flour. Mia makes $\frac{1}{2}$ of the recipe.

How much flour does she use?

Flour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 3 • FORWARD 1 • Q5/5

92

QUESTION ID

S3-F1-Q5

TYPE

Short answer

CORRECT ANSWER

1 1/2 cups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 3 • FORWARD 1 • Q5/5

SET 3 • 4.OA.A.2

FORWARD 2

Solve unit rate problems • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

5 notebooks cost \$15.

At the same price, how much do 8 notebooks cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 3 • FORWARD 2 • Q1/5

94

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

\$24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 3 • FORWARD 2 • Q1/5

True or false?

If 4 pounds of grapes cost \$10, then 1 pound costs \$2.50.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 3 • FORWARD 2 • Q2/5

95

QUESTION ID

S3-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 3 • FORWARD 2 • Q2/5

A car goes 180 miles in 3 hours at a steady speed.

How far does it go in 5 hours?

- A.** 240 miles
- B.** 360 miles
- C.** 300 miles
- D.** 185 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 3 • FORWARD 2 • Q3/5

96

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 300 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 3 • FORWARD 2 • Q3/5

A printer prints 42 pages in 6 minutes.

How many pages does it print per minute?

Pages per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 3 • FORWARD 2 • Q4/5

97

QUESTION ID

S3-F2-Q4

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 3 • FORWARD 2 • Q4/5

True or false?

If 3 tickets cost \$27, then 7 tickets cost \$56.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 3 • FORWARD 2 • Q5/5

98

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 3 • FORWARD 2 • Q5/5

Solve multiplicative comparison word problems by dividing

MAIN	Grade 4 main question	4.OA.A.2
BACKWARD 1	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 2	Find the unknown number in a multiplication or division equation	3.OA.A.4
FORWARD 1	Solve division word problems with two-digit divisors	5.NBT.B.6
FORWARD 2	Solve one-step equations of the form $px = q$	6.EE.B.7

SET 4 • 4.OA.A.2

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A dog weighs 48 pounds. That is 6 times as much as a cat weighs.

How much does the cat weigh?

Weight: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 4 • MAIN • Q1/5

101

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

8 pounds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 4 • MAIN • Q1/5

Leo has 72 cards. That is 9 times as many cards as Ty has.

How many cards does Ty have?

Cards: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 4 • MAIN • Q2/5

102

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 4 • MAIN • Q2/5

A building is 60 feet tall. That is 5 times as tall as a tree.

How tall is the tree?

- A. 300 feet
- B. 55 feet
- C. 65 feet
- D. 12 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 4 • MAIN • Q3/5

103

QUESTION ID

S4-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 4 • MAIN • Q3/5

A jacket costs \$56. A hat costs \$8.

The jacket costs how many times as much as the hat?

Times as much: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 4 • MAIN • Q4/5

104

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 4 • MAIN • Q4/5

True or false?

Mo ran 36 laps. That is 4 times as many laps as Ed ran. Ed ran 9 laps.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 4 • MAIN • Q5/5

105

QUESTION ID

S4-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 4 • MAIN • Q5/5

SET 4 • 4.OA.A.2

BACKWARD 1

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $42 \div 6$ by thinking $6 \times ? = 42$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 4 • BACKWARD 1 • Q1/5

107

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 4 • BACKWARD 1 • Q1/5

True or false?

$54 \div 9 = 6$ because $9 \times 6 = 54$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 4 • BACKWARD 1 • Q2/5

108

QUESTION ID

S4-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 4 • BACKWARD 1 • Q2/5

Which fact helps you find $63 \div 7$?

- A. $9 + 7 = 16$
- B. $63 - 7 = 56$
- C. $7 \times 9 = 63$
- D. $7 \times 7 = 49$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 4 • BACKWARD 1 • Q3/5

109

QUESTION ID

S4-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $7 \times 9 = 63$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 4 • BACKWARD 1 • Q3/5

Divide.

$$32 \div 8$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 4 • BACKWARD 1 • Q4/5

110

QUESTION ID

S4-B1-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 4 • BACKWARD 1 • Q4/5

True or false?

$48 \div 6 = 9$ because $6 \times 9 = 48$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 4 • BACKWARD 1 • Q5/5

111

QUESTION ID

S4-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 4.OA.A.2

BACKWARD 2

Find the unknown number in a multiplication or division equation • Grade 3 •
Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

$$5 \times ? = 40$$

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 4 • BACKWARD 2 • Q1/5

113

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 4 • BACKWARD 2 • Q1/5

True or false?

If $27 \div ? = 3$, the unknown number is 9.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 4 • BACKWARD 2 • Q2/5

114

QUESTION ID

S4-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 4 • BACKWARD 2 • Q2/5

$$? \times 9 = 72$$

- A. 63
- B. 9
- C. 81
- D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 4 • BACKWARD 2 • Q3/5

115

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 4 • BACKWARD 2 • Q3/5

Find the unknown number.

$$? \div 4 = 6$$

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 4 • BACKWARD 2 • Q4/5

116

QUESTION ID

S4-B2-Q4

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 4 • BACKWARD 2 • Q4/5

True or false?

If $6 \times ? = 36$, the unknown number is 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 4 • BACKWARD 2 • Q5/5

117

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 4.OA.A.2

FORWARD 1

Solve division word problems with two-digit divisors • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

A truck carries 384 boxes. That is 12 times as many boxes as a van carries.

How many boxes does the van carry?

Boxes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 4 • FORWARD 1 • Q1/5

119

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 4 • FORWARD 1 • Q1/5

A school has 864 students in 24 classes of the same size.

How many students are in each class?

A. 38

B. 36

C. 34

D. 840

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 4 • FORWARD 1 • Q2/5

120

QUESTION ID

S4-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 4 • FORWARD 1 • Q2/5

True or false?

$$1,050 \div 15 = 70$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 4 • FORWARD 1 • Q3/5

121

QUESTION ID

S4-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 4 • FORWARD 1 • Q3/5

Divide.

$$2,016 \div 36$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 4 • FORWARD 1 • Q4/5

122

QUESTION ID

S4-F1-Q4

TYPE

Short answer

CORRECT ANSWER

56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 4 • FORWARD 1 • Q4/5

True or false?

A farm has 630 trees in rows of 18 trees. There are 45 rows.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 4 • FORWARD 1 • Q5/5

123

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 4 • FORWARD 1 • Q5/5

SET 4 • 4.OA.A.2

FORWARD 2

Solve one-step equations of the form $px = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for c.

$$6c = 48$$

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 4 • FORWARD 2 • Q1/5

125

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 4 • FORWARD 2 • Q1/5

True or false?

The solution of $9x = 72$ is $x = 8$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 4 • FORWARD 2 • Q2/5

126

QUESTION ID

S4-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 4 • FORWARD 2 • Q2/5

Solve for n.

$$5n = 65$$

- A.** 325
- B.** 60
- C.** 70
- D.** 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 4 • FORWARD 2 • Q3/5

127

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 4 • FORWARD 2 • Q3/5

A dog weighs 48 pounds, which is 6 times the weight c of a cat.

Write and solve an equation for c .

Equation: _____

$c =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 4 • FORWARD 2 • Q4/5

128

QUESTION ID

S4-F2-Q4

TYPE

Short answer

CORRECT ANSWER

**$6c = 48; c = 8$
pounds**

GRADING NOTE Any equivalent equation is correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 4 • FORWARD 2 • Q4/5

True or false?

The solution of $4y = 36$ is $y = 32$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 4 • FORWARD 2 • Q5/5

129

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 4 • FORWARD 2 • Q5/5

Tell multiplicative comparison from additive comparison

MAIN	Grade 4 main question	4.OA.A.2
BACKWARD 1	Compare quantities by adding or subtracting	2.OA.A.1
BACKWARD 2	Multiplication facts within 100	3.OA.C.7
FORWARD 1	Interpret numerical expressions without evaluating them	5.OA.A.2
FORWARD 2	Tell an additive relationship from a multiplicative one in tables a...	6.EE.C.9

SET 5 • 4.OA.A.2

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

Ana has 4 shells. Ben has 12 shells.

Which statement is true?

- A. Ben has 3 times as many shells as Ana.
- B. Ben has 3 more shells than Ana.
- C. Ana has 8 times as many shells as Ben.
- D. Ben has 4 more shells than Ana.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 5 • MAIN • Q1/5

132

QUESTION ID

S5-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

**A. Ben has 3 times
as many shells as
Ana.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 5 • MAIN • Q1/5

The red team scored 24 points. The blue team scored 6 points.

How many more points did the red team score? How many times as many points?

How many more: _____

How many times as many: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 5 • MAIN • Q2/5

133

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

18 more; 4 times as many

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 5 • MAIN • Q2/5

True or false?

"8 more than 5" is 40.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 5 • MAIN • Q3/5

134

QUESTION ID

S5-M-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 5 • MAIN • Q3/5

Which question is answered by multiplying?

- A.** A cat is 3 years old. A dog is 4 times as old. How old is the dog?
- B.** A cat ate 4 treats and then 3 more. How many did it eat?
- C.** A cat is 3 years old. A dog is 4 years older. How old is the dog?
- D.** A cat is 7 years old. A dog is 3 years old. How much older is the cat?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 5 • MAIN • Q4/5

135

QUESTION ID

S5-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. A cat is 3 years old. A dog is 4 times as old. How old is the dog?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 5 • MAIN • Q4/5

True or false?

A pencil is 6 inches long. A ruler that is 2 times as long is 12 inches, but a crayon that is 2 inches longer than the pencil is 8 inches.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 5 • MAIN • Q5/5

136

QUESTION ID

S5-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 5 • MAIN • Q5/5

SET 5 • 4.OA.A.2

BACKWARD 1

Compare quantities by adding or subtracting • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Jo has 15 stamps. Ty has 9 stamps.
How many more stamps does Jo have?

Stamps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q1/5

138

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q1/5

True or false?

Mia has 8 beads. That is 5 more beads than Kim has. Kim has 3 beads.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q2/5

139

QUESTION ID

S5-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q2/5

A red rope is 34 cm long. A green rope is 18 cm long.

How much longer is the red rope?

- A.** 14 cm
- B.** 16 cm
- C.** 52 cm
- D.** 26 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q3/5

140

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 16 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q3/5

A tower has 27 blocks. Another tower has 8 more blocks.

How many blocks does it have?

Blocks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q4/5

141

QUESTION ID

S5-B1-Q4

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q4/5

True or false?

Sam has 20 cards. Lee has 6 fewer cards. Lee has 26 cards.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q5/5

142

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 4.OA.A.2

BACKWARD 2

Multiplication facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$4 \times 9$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 5 • BACKWARD 2 • Q1/5

144

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 5 • BACKWARD 2 • Q1/5

True or false?

$$6 \times 7 = 42$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 5 • BACKWARD 2 • Q2/5

145

QUESTION ID

S5-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 5 • BACKWARD 2 • Q2/5

$$3 \times 8 = ?$$

- A.** 21
- B.** 24
- C.** 11
- D.** 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 5 • BACKWARD 2 • Q3/5

146

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 5 • BACKWARD 2 • Q3/5

Multiply.

$$5 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 5 • BACKWARD 2 • Q4/5

147

QUESTION ID

S5-B2-Q4

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 5 • BACKWARD 2 • Q4/5

True or false?

$$8 \times 8 = 16$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 5 • BACKWARD 2 • Q5/5

148

QUESTION ID

S5-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 4.OA.A.2

FORWARD 1

Interpret numerical expressions without evaluating them • Grade 5 •
Operations & Algebraic Thinking • 5.OA.A.2

True or false?

$6 \times (25 + 9)$ is 6 times as large as $25 + 9$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • FORWARD 1 • Q1/5

150

QUESTION ID

S5-F1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • FORWARD 1 • Q1/5

How does $4 \times (17 - 3)$ compare with $17 - 3$?

- A.** It is 4 times as large.
- B.** It is the same.
- C.** It is 4 less.
- D.** It is 4 more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • FORWARD 1 • Q2/5

151

QUESTION ID

S5-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. It is 4 times as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • FORWARD 1 • Q2/5

True or false?

$(48 + 6) + 3$ is 3 times as large as $48 + 6$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • FORWARD 1 • Q3/5

152

QUESTION ID

S5-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • FORWARD 1 • Q3/5

Without calculating, how does $10 \times (325 + 19)$ compare with $325 + 19$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • FORWARD 1 • Q4/5

153

QUESTION ID

S5-F1-Q4

TYPE

Short answer

CORRECT ANSWER

It is 10 times as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • FORWARD 1 • Q4/5

Which expression is 2 more than 9×7 ?

- A.** $9 \times (7 + 2)$
- B.** $2 \times 9 \times 7$
- C.** $9 \times 7 + 2$
- D.** $(9 + 2) \times 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • FORWARD 1 • Q5/5

154

QUESTION ID

S5-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. $9 \times 7 + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • FORWARD 1 • Q5/5

SET 5 • 4.OA.A.2

FORWARD 2

Tell an additive relationship from a multiplicative one in tables and equations •

Grade 6 • Expressions & Equations • 6.EE.C.9

Which equation shows that y is always 4 times x ?

- A. $y = 4x$
- B. $y = x + 4$
- C. $y = x - 4$
- D. $x = 4y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • FORWARD 2 • Q1/5

156

QUESTION ID

S5-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. $y = 4x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • FORWARD 2 • Q1/5

True or false?

The table fits the equation $y = 5x$.

x	1	2	3
y	5	6	7

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • FORWARD 2 • Q2/5

157

QUESTION ID

S5-F2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • FORWARD 2 • Q2/5

Write an equation for y in terms of x .

x	2	4	6
y	6	12	18

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • FORWARD 2 • Q3/5

158

QUESTION ID

S5-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$y = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • FORWARD 2 • Q3/5

Write an equation for y in terms of x .

x	2	4	6
y	5	7	9

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • FORWARD 2 • Q4/5

159

QUESTION ID

S5-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$$y = x + 3$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • FORWARD 2 • Q4/5

True or false?

In $y = x + 6$, y is always 6 more than x .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • FORWARD 2 • Q5/5

160

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • FORWARD 2 • Q5/5

Solve multistep word problems and write an equation with a letter for the unknown

MAIN	Grade 4 main question	4.OA.A.3
BACKWARD 1	Two-step word problems	3.OA.D.8
BACKWARD 2	Add and subtract within 1000	3.NBT.A.2
FORWARD 1	Evaluate expressions with parentheses, brackets, and braces	5.OA.A.1
FORWARD 2	Use variables to write expressions for real-world problems	6.EE.B.6

SET 6 • 4.OA.A.3

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

A school bought 8 packs of 24 pencils. It gave away 57 pencils.

Write an equation with a letter for the number of pencils left. Then solve it.

Equation: _____

Pencils left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 6 • MAIN • Q1/5

163

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

$$p = 8 \times 24 - 57;$$

$$p = 135$$

GRADING NOTE Any equivalent equation with a letter is correct. $8 \times 24 = 192$ and $192 - 57 = 135$. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 6 • MAIN • Q1/5

Ty has \$250. He buys 3 games that cost \$45 each.

Write an equation with a letter for the money left. Then solve it.

Equation: _____

Money left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 6 • MAIN • Q2/5

164

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

$$m = 250 - 3 \times 45;$$

$$m = \$115$$

GRADING NOTE Any equivalent equation with a letter is correct. $3 \times 45 = 135$ and $250 - 135 = 115$. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 6 • MAIN • Q2/5

A farmer has 6 crates with 35 apples in each crate. She sells 98 apples.

Which equation gives the number of apples left, a ?

A. $a = 6 \times 35 - 98$

B. $a = 6 + 35 - 98$

C. $a = 98 - 6 \times 35$

D. $a = 6 \times (35 - 98)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 6 • MAIN • Q3/5

165

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $a = 6 \times 35 - 98$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 6 • MAIN • Q3/5

Tickets cost \$9 for adults and \$6 for children. A group buys 4 adult tickets and 7 child tickets.

Write an equation with a letter for the total cost. Then solve it.

Equation: _____

Total cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 6 • MAIN • Q4/5

166

QUESTION ID

S6-M-Q4

TYPE

Short answer

CORRECT ANSWER

$$c = 4 \times 9 + 7 \times 6;$$

$$c = \$78$$

GRADING NOTE Any equivalent equation with a letter is correct. $36 + 42 = 78$. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 6 • MAIN • Q4/5

True or false?

A baker makes 128 rolls and puts them in bags of 8. She sells 11 bags. The equation $b = 128 \div 8 - 11$ gives the bags left, and $b = 5$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 6 • MAIN • Q5/5

167

QUESTION ID

S6-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 6 • MAIN • Q5/5

SET 6 • 4.OA.A.3

BACKWARD 1

Two-step word problems • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

Ann had 45 beads. She bought 3 packs with 8 beads in each pack.

How many beads does she have now?

Beads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 6 • BACKWARD 1 • Q1/5

169

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

69

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 6 • BACKWARD 1 • Q1/5

True or false?

Jo has 50 cents. She spends 4 groups of 5 cents.
She has 30 cents left.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 6 • BACKWARD 1 • Q2/5

170

QUESTION ID

S6-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 6 • BACKWARD 1 • Q2/5

There are 7 rows of 6 chairs. 15 chairs are taken.

How many chairs are empty?

- A. 13
- B. 42
- C. 57
- D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 6 • BACKWARD 1 • Q3/5

171

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 6 • BACKWARD 1 • Q3/5

Leo has 3 boxes of 9 markers. He gives away 8 markers.
How many markers does he have left?

Markers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 6 • BACKWARD 1 • Q4/5

172

QUESTION ID

S6-B1-Q4

TYPE

Short answer

CORRECT ANSWER

19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 6 • BACKWARD 1 • Q4/5

True or false?

6 bags of 9 oranges, with 10 oranges eaten,
leaves 50 oranges.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 6 • BACKWARD 1 • Q5/5

173

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 4.OA.A.3

BACKWARD 2

Add and subtract within 1000 • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.2

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$468 + 257$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 6 • BACKWARD 2 • Q1/5

175

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

725

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 6 • BACKWARD 2 • Q1/5

175

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$803 - 469$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 6 • BACKWARD 2 • Q2/5

176

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

334

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 6 • BACKWARD 2 • Q2/5

176

True or false?

$$350 + 475 = 825$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 6 • BACKWARD 2 • Q3/5

177

QUESTION ID

S6-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 6 • BACKWARD 2 • Q3/5

$$612 - 238 = ?$$

- A.** 484
- B.** 374
- C.** 386
- D.** 850

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 6 • BACKWARD 2 • Q4/5

178

QUESTION ID

S6-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 374

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 6 • BACKWARD 2 • Q4/5

True or false?

$$900 - 345 = 665$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 6 • BACKWARD 2 • Q5/5

179

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 4.OA.A.3

FORWARD 1

Evaluate expressions with parentheses, brackets, and braces • Grade 5 •
Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$8 \times 24 - 57$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 6 • FORWARD 1 • Q1/5

181

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

135

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 6 • FORWARD 1 • Q1/5

Evaluate.

$$250 - 3 \times 45$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 6 • FORWARD 1 • Q2/5

182

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

115

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 6 • FORWARD 1 • Q2/5

Evaluate.

$$(36 + 12) \div 6$$

- A. 38
- B. 8
- C. 12
- D. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 6 • FORWARD 1 • Q3/5

183

QUESTION ID

S6-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 6 • FORWARD 1 • Q3/5

True or false?

$$4 \times (9 + 6) = 42$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 6 • FORWARD 1 • Q4/5

184

QUESTION ID

S6-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 6 • FORWARD 1 • Q4/5

Evaluate.

$$[40 - (5 \times 6)] \times 3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 6 • FORWARD 1 • Q5/5

185

QUESTION ID

S6-F1-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 4.OA.A.3

FORWARD 2

Use variables to write expressions for real-world problems • Grade 6 •
Expressions & Equations • 6.EE.B.6

Pens cost \$3 each.

Write an expression for the cost of p pens.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 6 • FORWARD 2 • Q1/5

187

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3p

GRADING NOTE $3 \times p$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 6 • FORWARD 2 • Q1/5

Ty has 40 stickers and gives away s stickers.

Which expression shows how many he has left?

A. $40 - s$

B. $s - 40$

C. $40s$

D. $40 + s$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 6 • FORWARD 2 • Q2/5

188

QUESTION ID

S6-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $40 - s$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 6 • FORWARD 2 • Q2/5

A movie ticket costs \$9 and a popcorn costs \$4.

Write an expression for the cost of t tickets and one popcorn.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 6 • FORWARD 2 • Q3/5

189

QUESTION ID

S6-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$9t + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 6 • FORWARD 2 • Q3/5

True or false?

If n is the number of packs with 12 cards in each pack, then $12n$ is the total number of cards.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 6 • FORWARD 2 • Q4/5

190

QUESTION ID

S6-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 6 • FORWARD 2 • Q4/5

True or false?

The expression $5 + b$ means 5 times b .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 6 • FORWARD 2 • Q5/5

191

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 6 • FORWARD 2 • Q5/5

Interpret remainders in word problems

MAIN	Grade 4 main question	4.OA.A.3
BACKWARD 1	Quotients as shares or as groups	3.OA.A.2
BACKWARD 2	Division facts within 100	3.OA.C.7
FORWARD 1	Interpret a fraction as division to write a remainder as a fraction	5.NF.B.3
FORWARD 2	Divide multi-digit numbers with the standard algorithm	6.NS.B.2

SET 7 • 4.OA.A.3

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

29 students ride in cars that hold 4 students each.

How many cars are needed?

Cars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 7 • MAIN • Q1/5

194

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE $29 \div 4 = 7 \text{ R } 1$. The 1 extra student needs another car, so 8 cars.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 7 • MAIN • Q1/5

A baker has 50 muffins. She puts 6 muffins in each box.
How many boxes can she fill completely?

Boxes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 7 • MAIN • Q2/5

ID S7-M-Q2

195

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE $50 \div 6 = 8 \text{ R } 2$. The 2 left over do not fill a box.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 7 • MAIN • Q2/5

ID S7-M-Q2

Ella shares 35 stickers equally among 4 friends.

How many stickers are left over?

- A. 8
- B. 3
- C. 4
- D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 7 • MAIN • Q3/5

196

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 7 • MAIN • Q3/5

A rope 46 feet long is cut into 5-foot pieces.

How many full pieces are there, and how much rope is left over?

Full pieces: _____

Left over: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 7 • MAIN • Q4/5

197

QUESTION ID

S7-M-Q4

TYPE

Short answer

CORRECT ANSWER

9 pieces; 1 foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 7 • MAIN • Q4/5

True or false?

70 people need tables that seat 8 people each.
They need 8 tables.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 7 • MAIN • Q5/5

198

QUESTION ID

S7-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 7 • MAIN • Q5/5

SET 7 • 4.OA.A.3

BACKWARD 1

Quotients as shares or as groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

24 pencils are shared equally by 6 students.

How many pencils does each student get?

Pencils: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 1 • Q1/5

200

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 1 • Q1/5

True or false?

$30 \div 5$ can mean 30 objects put into groups of 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 1 • Q2/5

201

QUESTION ID

S7-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 1 • Q2/5

Which question does $18 \div 3$ answer?

- A. What is 3 more than 18?
- B. How many groups of 3 are in 18?
- C. What is 18 take away 3?
- D. How many are in 18 groups of 3?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 1 • Q3/5

202

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

**B. How many groups
of 3 are in 18?**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 1 • Q3/5

40 crayons are put into boxes of 8.

How many boxes are filled?

Boxes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 1 • Q4/5

203

QUESTION ID

S7-B1-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 1 • Q4/5

True or false?

12 ÷ 4 means 12 groups of 4.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 1 • Q5/5

204

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 4.OA.A.3

BACKWARD 2

Division facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Divide.

$$56 \div 7$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 7 • BACKWARD 2 • Q1/5

206

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 7 • BACKWARD 2 • Q1/5

True or false?

$$45 \div 5 = 9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 7 • BACKWARD 2 • Q2/5

207

QUESTION ID

S7-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 7 • BACKWARD 2 • Q2/5

$$64 \div 8 = ?$$

- A.** 56
- B.** 7
- C.** 9
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 7 • BACKWARD 2 • Q3/5

208

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 7 • BACKWARD 2 • Q3/5

Divide.

$$36 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 7 • BACKWARD 2 • Q4/5

209

QUESTION ID

S7-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 7 • BACKWARD 2 • Q4/5

True or false?

$$81 \div 9 = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 7 • BACKWARD 2 • Q5/5

210

QUESTION ID

S7-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 4.OA.A.3

FORWARD 1

Interpret a fraction as division to write a remainder as a fraction • Grade 5 •
Number & Operations–Fractions • 5.NF.B.3

3 friends share 7 pizzas equally.

How much pizza does each friend get? Write a mixed number.

Pizza: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 7 • FORWARD 1 • Q1/5

212

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2 $\frac{1}{3}$ pizzas

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 7 • FORWARD 1 • Q1/5

True or false?

When 9 feet of ribbon is cut into 4 equal pieces, each piece is $2\frac{1}{4}$ feet long.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 7 • FORWARD 1 • Q2/5

213

QUESTION ID

S7-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 7 • FORWARD 1 • Q2/5

5 sandwiches are shared equally by 2 people.

How much does each person get?

- A. $\frac{2}{5}$ sandwich
- B. 2 sandwiches
- C. $2\frac{1}{2}$ sandwiches
- D. 3 sandwiches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 7 • FORWARD 1 • Q3/5

214

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 1/2 sandwiches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 7 • FORWARD 1 • Q3/5

Write $17 \div 5$ as a mixed number.

Mixed number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 7 • FORWARD 1 • Q4/5

215

QUESTION ID

S7-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 7 • FORWARD 1 • Q4/5

True or false?

$$11 \div 4 = 2\frac{1}{4}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 7 • FORWARD 1 • Q5/5

216

QUESTION ID

S7-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 7 • FORWARD 1 • Q5/5

SET 7 • 4.OA.A.3

FORWARD 2

Divide multi-digit numbers with the standard algorithm • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$4,368 \div 52$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 7 • FORWARD 2 • Q1/5

218

QUESTION ID

S7-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

84

GRADING NOTE REQUIRED WORK: The standard division algorithm with every step shown (divide, multiply, subtract, bring down). A correct quotient without the algorithm earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 7 • FORWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$7,695 \div 81$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 7 • FORWARD 2 • Q2/5

219

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

95

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 7 • FORWARD 2 • Q2/5

219

Divide.

$$9,072 \div 36$$

- A. 25
- B. 252
- C. 242
- D. 262

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 7 • FORWARD 2 • Q3/5

220

QUESTION ID

S7-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 252

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 7 • FORWARD 2 • Q3/5

Divide. Write the quotient and the remainder.

$$1,000 \div 23$$

Quotient: _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 7 • FORWARD 2 • Q4/5

221

QUESTION ID

S7-F2-Q4

TYPE

Short answer

CORRECT ANSWER

43 R 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 7 • FORWARD 2 • Q4/5

True or false?

$$3,240 \div 45 = 72$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 7 • FORWARD 2 • Q5/5

222

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 7 • FORWARD 2 • Q5/5

Use rounding and mental math to check whether an answer is reasonable

MAIN	Grade 4 main question	4.OA.A.3
BACKWARD 1	Round whole numbers to the nearest 10 or 100	3.NBT.A.1
BACKWARD 2	Check answers with rounding	3.OA.D.8
FORWARD 1	Round decimals to estimate	5.NBT.A.4
FORWARD 2	Use estimates to check computations with multi-digit decimals —...	6.NS.B.3

SET 8 • 4.OA.A.3

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

Round each number to the nearest hundred. Then estimate the sum.

$$512 + 289$$

Rounded numbers: _____

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 8 • MAIN • Q1/5

225

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

500 and 300; 800

GRADING NOTE Both parts are required. The exact sum, 801, is not the estimate asked for.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 8 • MAIN • Q1/5

Mia says $48 \times 6 = 188$.

Round 48 to the nearest ten and estimate 48×6 . Is her answer reasonable?

Estimate: _____

Reasonable? _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 8 • MAIN • Q2/5

226

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

300; no

GRADING NOTE $50 \times 6 = 300$, which is far from 188 (the exact product is 288). Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 8 • MAIN • Q2/5

Round each number to the nearest ten. Which is the estimate of $73 + 28 + 41$?

- A. 150
- B. 142
- C. 140
- D. 130

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 8 • MAIN • Q3/5

227

QUESTION ID

S8-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 140

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 8 • MAIN • Q3/5

A store had 1,215 shirts and sold 487 of them.

Round each number to the nearest hundred to estimate how many shirts are left.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 8 • MAIN • Q4/5

228

QUESTION ID

S8-M-Q4

TYPE

Short answer

CORRECT ANSWER

700

GRADING NOTE $1,200 - 500 = 700$. The exact answer, 728, is not the estimate asked for.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 8 • MAIN • Q4/5

True or false?

Rounding each number to the nearest hundred,
 $395 + 612$ is about 900.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 8 • MAIN • Q5/5

229

QUESTION ID

S8-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 8 • MAIN • Q5/5

SET 8 • 4.OA.A.3

BACKWARD 1

Round whole numbers to the nearest 10 or 100 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1

Round 67 to the nearest ten.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 8 • BACKWARD 1 • Q1/5

231

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 8 • BACKWARD 1 • Q1/5

True or false?

348 rounded to the nearest hundred is 400.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 8 • BACKWARD 1 • Q2/5

232

QUESTION ID

S8-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 8 • BACKWARD 1 • Q2/5

Round 452 to the nearest ten.

- A.** 400
- B.** 460
- C.** 500
- D.** 450

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 8 • BACKWARD 1 • Q3/5

233

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 450

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 8 • BACKWARD 1 • Q3/5

Round 815 to the nearest hundred.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 8 • BACKWARD 1 • Q4/5

234

QUESTION ID

S8-B1-Q4

TYPE

Short answer

CORRECT ANSWER

800

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 8 • BACKWARD 1 • Q4/5

True or false?

75 rounded to the nearest ten is 80.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 8 • BACKWARD 1 • Q5/5

235

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 4.OA.A.3

BACKWARD 2

Check answers with rounding • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

True or false?

Rounding each number to the nearest ten,
 $38 + 51$ is about 90.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 8 • BACKWARD 2 • Q1/5

237

QUESTION ID

S8-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 8 • BACKWARD 2 • Q1/5

Leo says $297 + 102 = 499$. Rounding to the nearest hundred gives $300 + 100 = 400$.

What does the estimate show?

- A.** An estimate cannot check a sum. **B.** His answer is too big.
- C.** His answer is correct. **D.** His answer is too small.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 8 • BACKWARD 2 • Q2/5

238

QUESTION ID

S8-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. His answer is too big.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 8 • BACKWARD 2 • Q2/5

Round each number to the nearest ten. Then estimate.

$$62 - 29$$

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 8 • BACKWARD 2 • Q3/5

239

QUESTION ID

S8-B2-Q3

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 8 • BACKWARD 2 • Q3/5

True or false?

Rounding each number to the nearest hundred,
710 – 290 is about 600.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 8 • BACKWARD 2 • Q4/5

240

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 8 • BACKWARD 2 • Q4/5

Round each number to the nearest ten. Then estimate.

$$47 + 33$$

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 8 • BACKWARD 2 • Q5/5

241

QUESTION ID

S8-B2-Q5

TYPE

Short answer

CORRECT ANSWER

80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 4.OA.A.3

FORWARD 1

Round decimals to estimate • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round each number to the nearest whole number. Then estimate.

$$4.8 + 7.3$$

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 8 • FORWARD 1 • Q1/5

243

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 8 • FORWARD 1 • Q1/5

True or false?

Rounding each number to the nearest whole number, $9.6 - 3.2$ is about 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 8 • FORWARD 1 • Q2/5

244

QUESTION ID

S8-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 8 • FORWARD 1 • Q2/5

Round 6.47 to the nearest tenth.

- A.** 6.4
- B.** 6
- C.** 6.5
- D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 8 • FORWARD 1 • Q3/5

245

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 8 • FORWARD 1 • Q3/5

Round each number to the nearest whole number. Then estimate.

$$2.9 \times 4.1$$

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 8 • FORWARD 1 • Q4/5

246

QUESTION ID

S8-F1-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 8 • FORWARD 1 • Q4/5

True or false?

Rounding each number to the nearest whole number, $12.5 + 6.5$ is about 18.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 8 • FORWARD 1 • Q5/5

247

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 8 • FORWARD 1 • Q5/5

FORWARD 2

Use estimates to check computations with multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

Round each number to the nearest whole number. Use your estimate to decide whether $18.75 + 6.4 = 25.15$ is reasonable.

Estimate: _____

Reasonable? _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 2 • Q1/5 • NEAREST RELATED

249

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

25; yes

GRADING NOTE $19 + 6 = 25$. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 2 • Q1/5 • NEAREST RELATED

True or false?

Rounding each number to the nearest whole number, 3.9×5.2 is about 20, so a product of 202.8 is not reasonable.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 2 • Q2/5 • NEAREST RELATED

250

QUESTION ID

S8-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 2 • Q2/5 • NEAREST RELATED

Ann computed $45.6 \div 3.8 = 1.2$. Rounding each number to the nearest whole number gives $46 \div 4$, which is about 11.

What does this show?

- A.** Her answer is too large. **B.** The estimate cannot be used. **C.** Her answer is far too small. **D.** Her answer is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 2 • Q3/5 • NEAREST RELATED

251

QUESTION ID

S8-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Her answer is far too small.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 2 • Q3/5 • NEAREST RELATED

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$12.4 \times 3.5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 2 • Q4/5 • NEAREST RELATED

252

QUESTION ID

S8-F2-Q4

TYPE

Short answer

CORRECT ANSWER

43.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 2 • Q4/5 • NEAREST RELATED

252

True or false?

Rounding each number to the nearest whole number, 7.1×2.9 is about 21, so a product of 2.059 is reasonable.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 2 • Q5/5 • NEAREST RELATED

253

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 2 • Q5/5 • NEAREST RELATED

Find all factor pairs of a whole number from 1 to 100

MAIN	Grade 4 main question	4.OA.B.4
BACKWARD 1	Find an unknown factor	3.OA.B.6
BACKWARD 2	Arrays with up to 5 rows and 5 columns	2.OA.C.4
FORWARD 1	Use multiples to find a common denominator	5.NF.A.1
FORWARD 2	Find the greatest common factor	6.NS.B.4

SET 9 • 4.OA.B.4

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 24.

Factor pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 9 • MAIN • Q1/5

256

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

**1 × 24, 2 × 12,
3 × 8, 4 × 6**

GRADING NOTE All four pairs are required, in any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 9 • MAIN • Q1/5

List all the factor pairs of 45.

Factor pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 9 • MAIN • Q2/5

257

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

1 × 45, 3 × 15, 5 × 9

GRADING NOTE All three pairs are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 9 • MAIN • Q2/5

Which is NOT a factor pair of 60?

- A.** 7×9
- B.** 5×12
- C.** 6×10
- D.** 4×15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 9 • MAIN • Q3/5

258

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7×9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 9 • MAIN • Q3/5

True or false?

36 has exactly 5 factor pairs.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 9 • MAIN • Q4/5

259

QUESTION ID

S9-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 9 • MAIN • Q4/5

How many factors does 18 have? List them.

How many: _____

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 9 • MAIN • Q5/5

260

QUESTION ID

S9-M-Q5

TYPE

Short answer

CORRECT ANSWER

6; 1, 2, 3, 6, 9, 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 9 • MAIN • Q5/5

SET 9 • 4.OA.B.4

BACKWARD 1

Find an unknown factor • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find the unknown factor.

$$4 \times ? = 24$$

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 9 • BACKWARD 1 • Q1/5

262

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 9 • BACKWARD 1 • Q1/5

True or false?

$3 \times 8 = 24$, so 3 and 8 are a factor pair of 24.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 9 • BACKWARD 1 • Q2/5

263

QUESTION ID

S9-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 9 • BACKWARD 1 • Q2/5

Which number times 5 equals 45?

- A.** 9
- B.** 50
- C.** 40
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 9 • BACKWARD 1 • Q3/5

264

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 9 • BACKWARD 1 • Q3/5

Find the unknown factor.

$$? \times 7 = 28$$

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 9 • BACKWARD 1 • Q4/5

265

QUESTION ID

S9-B1-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$$6 \times 7 = 48$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 9 • BACKWARD 1 • Q5/5

266

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 4.OA.B.4

BACKWARD 2

Arrays with up to 5 rows and 5 columns • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

How many dots are in the array?

Dots: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 9 • BACKWARD 2 • Q1/5

268

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 9 • BACKWARD 2 • Q1/5

True or false?

An array with 4 rows of 5 dots has 20 dots.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 9 • BACKWARD 2 • Q2/5

269

QUESTION ID

S9-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 9 • BACKWARD 2 • Q2/5

Which addition equation matches an array with 2 rows of 4?

A. $4 - 2 = 2$

B. $2 + 2 = 4$

C. $4 + 4 = 8$

D. $2 + 4 = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 9 • BACKWARD 2 • Q3/5

270

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $4 + 4 = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 9 • BACKWARD 2 • Q3/5

Write an addition equation for the array.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 9 • BACKWARD 2 • Q4/5

271

QUESTION ID

S9-B2-Q4

TYPE

Short answer

CORRECT ANSWER

$$4 + 4 + 4 = 12$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 9 • BACKWARD 2 • Q4/5

True or false?

An array with 5 rows of 3 dots has 8 dots.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 9 • BACKWARD 2 • Q5/5

272

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 4.OA.B.4

FORWARD 1

Use multiples to find a common denominator • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Find the least common denominator of $\frac{1}{4}$ and $\frac{1}{6}$.

Denominator: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 9 • FORWARD 1 • Q1/5

274

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 9 • FORWARD 1 • Q1/5

True or false?

$\frac{2}{3}$ and $\frac{3}{4}$ can both be written with a denominator of 12.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 9 • FORWARD 1 • Q2/5

275

QUESTION ID

S9-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 9 • FORWARD 1 • Q2/5

Which is a common denominator of $\frac{1}{3}$ and $\frac{2}{5}$?

- A. 3
- B. 15
- C. 10
- D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 9 • FORWARD 1 • Q3/5

276

QUESTION ID

S9-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 9 • FORWARD 1 • Q3/5

Add.

$$\frac{1}{2} + \frac{1}{3}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 9 • FORWARD 1 • Q4/5

277

QUESTION ID

S9-F1-Q4

TYPE

Short answer

CORRECT ANSWER

5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 9 • FORWARD 1 • Q4/5

True or false?

A common denominator of $\frac{1}{4}$ and $\frac{1}{10}$ is 14.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 9 • FORWARD 1 • Q5/5

278

QUESTION ID

S9-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 9 • FORWARD 1 • Q5/5

SET 9 • 4.OA.B.4

FORWARD 2

Find the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 9 • FORWARD 2 • Q1/5

280

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 9 • FORWARD 2 • Q1/5

Find the greatest common factor of 18 and 45.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 9 • FORWARD 2 • Q2/5

281

QUESTION ID

S9-F2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 9 • FORWARD 2 • Q2/5

What is the greatest common factor of 16 and 40?

- A.** 2
- B.** 16
- C.** 4
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 9 • FORWARD 2 • Q3/5

282

QUESTION ID

S9-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 9 • FORWARD 2 • Q3/5

True or false?

The greatest common factor of 14 and 35 is 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 9 • FORWARD 2 • Q4/5

283

QUESTION ID

S9-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 9 • FORWARD 2 • Q4/5

True or false?

The greatest common factor of 20 and 30 is 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 9 • FORWARD 2 • Q5/5

284

QUESTION ID

S9-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 9 • FORWARD 2 • Q5/5

A whole number is a multiple of each of its factors; find multiples

MAIN	Grade 4 main question	4.OA.B.4
BACKWARD 1	Skip count and find patterns in the multiplication table	3.OA.D.9
BACKWARD 2	Odd and even numbers	2.OA.C.3
FORWARD 1	Generate two patterns and compare their terms	5.OA.B.3
FORWARD 2	Find the least common multiple	6.NS.B.4

SET 10 • 4.OA.B.4

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first six positive multiples of 7.

Multiples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 10 • MAIN • Q1/5

287

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

7, 14, 21, 28, 35, 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 10 • MAIN • Q1/5

True or false?

Because 6 is a factor of 42, 42 is a multiple of 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 10 • MAIN • Q2/5

288

QUESTION ID

S10-M-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 10 • MAIN • Q2/5

Which number is a multiple of 9?

- A.** 39
- B.** 19
- C.** 54
- D.** 92

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 10 • MAIN • Q3/5

289

QUESTION ID

S10-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 10 • MAIN • Q3/5

Is 84 a multiple of 6? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 10 • MAIN • Q4/5

290

QUESTION ID

S10-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE $6 \times 14 = 84$ (or $84 \div 6 = 14$ with no remainder), so 6 is a factor of 84 and 84 is a multiple of 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 10 • MAIN • Q4/5

True or false?

70 is a multiple of 4.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 10 • MAIN • Q5/5

291

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 10 • MAIN • Q5/5

SET 10 • 4.OA.B.4

BACKWARD 1

Skip count and find patterns in the multiplication table • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

Skip count by 6.

6, 12, 18, __, __

Next two: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 10 • BACKWARD 1 • Q1/5

293

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

24, 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 10 • BACKWARD 1 • Q1/5

True or false?

Every number in the 5s row of the multiplication table ends in 0 or 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 10 • BACKWARD 1 • Q2/5

294

QUESTION ID

S10-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 10 • BACKWARD 1 • Q2/5

Which number is in the 4s row of the multiplication table?

- A. 18
- B. 30
- C. 28
- D. 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 10 • BACKWARD 1 • Q3/5

295

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 10 • BACKWARD 1 • Q3/5

What comes next?

9, 18, 27, 36, ____

Next: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 10 • BACKWARD 1 • Q4/5

296

QUESTION ID

S10-B1-Q4

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 10 • BACKWARD 1 • Q4/5

True or false?

Every number in the 3s row of the multiplication table is odd.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 10 • BACKWARD 1 • Q5/5

297

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 4.OA.B.4

BACKWARD 2

Odd and even numbers • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

True or false?

16 is even.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 10 • BACKWARD 2 • Q1/5

299

QUESTION ID

S10-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 10 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Is 23 odd or even?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 10 • BACKWARD 2 • Q2/5

300

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Odd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 10 • BACKWARD 2 • Q2/5

Which number is even?

- A.** 27
- B.** 38
- C.** 41
- D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 10 • BACKWARD 2 • Q3/5

301

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 38

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 10 • BACKWARD 2 • Q3/5

Write 14 as the sum of two equal addends.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 10 • BACKWARD 2 • Q4/5

302

QUESTION ID

S10-B2-Q4

TYPE

Short answer

CORRECT ANSWER

7 + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 10 • BACKWARD 2 • Q4/5

True or false?

An even number of objects cannot be put into two equal groups.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 10 • BACKWARD 2 • Q5/5

303

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 4.OA.B.4

FORWARD 1

Generate two patterns and compare their terms • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 12.

Write the first five terms of each pattern.

Pattern A: _____

Pattern B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 10 • FORWARD 1 • Q1/5

305

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Pattern A: 0, 4, 8, 12, 16; Pattern B: 0, 12, 24, 36, 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 10 • FORWARD 1 • Q1/5

True or false?

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 12. Each term of B is 3 times the matching term of A.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 10 • FORWARD 1 • Q2/5

306

QUESTION ID

S10-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 10 • FORWARD 1 • Q2/5

Pattern A: 0, 5, 10, 15. Pattern B: 0, 10, 20, 30.

Which ordered pair (A, B) comes from the patterns?

- A. (15, 30)
- B. (15, 20)
- C. (30, 15)
- D. (10, 30)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 10 • FORWARD 1 • Q3/5

307

QUESTION ID

S10-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (15, 30)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 10 • FORWARD 1 • Q3/5

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 10.

What is the 5th term of Pattern B?

Term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 10 • FORWARD 1 • Q4/5

308

QUESTION ID

S10-F1-Q4

TYPE

Short answer

CORRECT ANSWER

40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 10 • FORWARD 1 • Q4/5

True or false?

Each term of Pattern B is 4 times the matching term of Pattern A. When A is 9, B is 13.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 10 • FORWARD 1 • Q5/5

309

QUESTION ID

S10-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 4.OA.B.4

FORWARD 2

Find the least common multiple • Grade 6 • The Number System • 6.NS.B.4

Find the least common multiple of 6 and 8.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 10 • FORWARD 2 • Q1/5

311

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 10 • FORWARD 2 • Q1/5

Find the least common multiple of 5 and 7.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 10 • FORWARD 2 • Q2/5

312

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 10 • FORWARD 2 • Q2/5

What is the least common multiple of 4 and 10?

- A.** 20
- B.** 2
- C.** 14
- D.** 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 10 • FORWARD 2 • Q3/5

313

QUESTION ID

S10-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 10 • FORWARD 2 • Q3/5

True or false?

The least common multiple of 9 and 12 is 36.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 10 • FORWARD 2 • Q4/5

314

QUESTION ID

S10-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 10 • FORWARD 2 • Q4/5

True or false?

The least common multiple of 3 and 6 is 18.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 10 • FORWARD 2 • Q5/5

315

QUESTION ID

S10-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 10 • FORWARD 2 • Q5/5

Decide whether a whole number from 1 to 100 is prime or composite

MAIN	Grade 4 main question	4.OA.B.4
BACKWARD 1	Find the unknown number in a multiplication or division equation	3.OA.A.4
BACKWARD 2	Arrays with up to 5 rows and 5 columns	2.OA.C.4
FORWARD 1	Evaluate expressions with grouping symbols	5.OA.A.1
FORWARD 2	Write and evaluate expressions with whole-number exponents	6.EE.A.1

SET 11 • 4.OA.B.4

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

Which number is prime?

- A.** 49
- B.** 33
- C.** 29
- D.** 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 11 • MAIN • Q1/5

318

QUESTION ID

S11-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 11 • MAIN • Q1/5

True or false?

51 is a prime number.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 11 • MAIN • Q2/5

319

QUESTION ID

S11-M-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 11 • MAIN • Q2/5

Is 39 prime or composite? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 11 • MAIN • Q3/5

320

QUESTION ID

S11-M-Q3

TYPE

Short answer

CORRECT ANSWER

Composite

GRADING NOTE $39 = 3 \times 13$, so 39 has factors other than 1 and itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 11 • MAIN • Q3/5

List all the prime numbers between 20 and 40.

Primes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 11 • MAIN • Q4/5

321

QUESTION ID

S11-M-Q4

TYPE

Short answer

CORRECT ANSWER

23, 29, 31, 37

GRADING NOTE All four are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 11 • MAIN • Q4/5

True or false?

A composite number has more than two factors.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 11 • MAIN • Q5/5

322

QUESTION ID

S11-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 11 • MAIN • Q5/5

SET 11 • 4.OA.B.4

BACKWARD 1

Find the unknown number in a multiplication or division equation • Grade 3 •
Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

$$? \times 3 = 21$$

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 1 • Q1/5

324

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 1 • Q1/5

True or false?

If $7 \times ? = 49$, the unknown number is 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 1 • Q2/5

325

QUESTION ID

S11-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 1 • Q2/5

$$9 \times ? = 54$$

- A.** 7
- B.** 45
- C.** 6
- D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 1 • Q3/5

326

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 1 • Q3/5

Find the unknown number.

$$35 \div ? = 5$$

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 1 • Q4/5

327

QUESTION ID

S11-B1-Q4

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 1 • Q4/5

True or false?

If $? \times 4 = 32$, the unknown number is 9.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 1 • Q5/5

328

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 4.OA.B.4

BACKWARD 2

Arrays with up to 5 rows and 5 columns • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

Write an addition equation for the array.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 11 • BACKWARD 2 • Q1/5

330

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5 + 5 = 10

GRADING NOTE $2 + 2 + 2 + 2 + 2 = 10$ (adding the columns) is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 11 • BACKWARD 2 • Q1/5

True or false?

5 dots can make a rectangle only as 1 row of 5 or 5 rows of 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 11 • BACKWARD 2 • Q2/5

331

QUESTION ID

S11-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 11 • BACKWARD 2 • Q2/5

Which array has 12 dots?

- A.** 3 rows of 4
- B.** 4 rows of 4
- C.** 3 rows of 5
- D.** 2 rows of 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 11 • BACKWARD 2 • Q3/5

332

QUESTION ID

S11-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3 rows of 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 11 • BACKWARD 2 • Q3/5

How many dots are in an array with 4 rows of 4?

Dots: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 11 • BACKWARD 2 • Q4/5

333

QUESTION ID

S11-B2-Q4

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 11 • BACKWARD 2 • Q4/5

True or false?

An array with 2 rows of 3 dots has 5 dots.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 11 • BACKWARD 2 • Q5/5

334

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 4.OA.B.4

FORWARD 1

Evaluate expressions with grouping symbols • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$2 \times 3 \times 7$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 11 • FORWARD 1 • Q1/5

336

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 11 • FORWARD 1 • Q1/5

Evaluate.

$$2 \times (5 \times 5)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 11 • FORWARD 1 • Q2/5

337

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 11 • FORWARD 1 • Q2/5

Evaluate.

$$(2 \times 2) \times (3 \times 3)$$

- A. 10
- B. 36
- C. 24
- D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 11 • FORWARD 1 • Q3/5

338

QUESTION ID

S11-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 11 • FORWARD 1 • Q3/5

True or false?

$$3 \times (2 \times 11) = 66$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 11 • FORWARD 1 • Q4/5

339

QUESTION ID

S11-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 11 • FORWARD 1 • Q4/5

True or false?

$$[2 \times (3 + 5)] \times 2 = 22$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 11 • FORWARD 1 • Q5/5

340

QUESTION ID

S11-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 11 • FORWARD 1 • Q5/5

SET 11 • 4.OA.B.4

FORWARD 2

Write and evaluate expressions with whole-number exponents • Grade 6 •
Expressions & Equations • 6.EE.A.1

Evaluate.

$$2^3 \times 3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 11 • FORWARD 2 • Q1/5

342

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 11 • FORWARD 2 • Q1/5

Write $2 \times 2 \times 2 \times 2 \times 5$ using an exponent.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 11 • FORWARD 2 • Q2/5

343

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$2^4 \times 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 11 • FORWARD 2 • Q2/5

Which expression equals 72?

- A. 6^2
- B. $2^2 \times 3^3$
- C. $2^3 \times 3^2$
- D. $2 \times 3 \times 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 11 • FORWARD 2 • Q3/5

344

QUESTION ID

S11-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $2^3 \times 3^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 11 • FORWARD 2 • Q3/5

True or false?

$$3^2 \times 5 = 45$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 11 • FORWARD 2 • Q4/5

345

QUESTION ID

S11-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 11 • FORWARD 2 • Q4/5

True or false?

$$2^5 = 10$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 11 • FORWARD 2 • Q5/5

346

QUESTION ID

S11-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 11 • FORWARD 2 • Q5/5

Generate a number or shape pattern that follows a given rule

MAIN	Grade 4 main question	4.OA.C.5
BACKWARD 1	Arithmetic patterns	3.OA.D.9
BACKWARD 2	Add or subtract 10 or 100 mentally	2.NBT.B.8
FORWARD 1	Form ordered pairs from two patterns	5.OA.B.3
FORWARD 2	Write an equation for a pattern with dependent and independent v...	6.EE.C.9

SET 12 • 4.OA.C.5

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "add 7." The pattern starts at 4.

Write the first five terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 12 • MAIN • Q1/5

349

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

4, 11, 18, 25, 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 12 • MAIN • Q1/5

The rule is "multiply by 3." The pattern starts at 2.

Write the first four terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 12 • MAIN • Q2/5

350

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

2, 6, 18, 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 12 • MAIN • Q2/5

Which pattern follows the rule "subtract 6," starting at 50?

- A. 50, 45, 40, 35
- B. 50, 44, 38, 32
- C. 50, 56, 62, 68
- D. 44, 38, 32, 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 12 • MAIN • Q3/5

351

QUESTION ID

S12-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 50, 44, 38, 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 12 • MAIN • Q3/5

True or false?

The rule "add 9," starting at 15, gives 15, 24, 33, 43.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 12 • MAIN • Q4/5

352

QUESTION ID

S12-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 12 • MAIN • Q4/5

In a shape pattern, each figure has 2 more squares than the one before. The 1st figure has 3 squares.

How many squares are in the 6th figure?

Squares: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 12 • MAIN • Q5/5

353

QUESTION ID

S12-M-Q5

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE 3, 5, 7, 9, 11, 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 12 • MAIN • Q5/5

SET 12 • 4.OA.C.5

BACKWARD 1

Arithmetic patterns • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

What comes next?

12, 16, 20, 24, ____

Next: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 12 • BACKWARD 1 • Q1/5

355

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 12 • BACKWARD 1 • Q1/5

True or false?

The pattern 5, 10, 15, 20 adds 5 each time.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 12 • BACKWARD 1 • Q2/5

356

QUESTION ID

S12-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 12 • BACKWARD 1 • Q2/5

What is the rule for 30, 27, 24, 21?

- A.** Add 3
- B.** Multiply by 3
- C.** Subtract 3
- D.** Subtract 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 12 • BACKWARD 1 • Q3/5

357

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Subtract 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 12 • BACKWARD 1 • Q3/5

What comes next?

7, 14, 21, 28, ____

Next: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 12 • BACKWARD 1 • Q4/5

358

QUESTION ID

S12-B1-Q4

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 12 • BACKWARD 1 • Q4/5

True or false?

The pattern 2, 4, 8, 16 adds 2 each time.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 12 • BACKWARD 1 • Q5/5

359

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 4.OA.C.5

BACKWARD 2

Add or subtract 10 or 100 mentally • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8

ANSWER KEY

Reduced copy of the question page, with its answer

What is 10 more than 347?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 12 • BACKWARD 2 • Q1/5

361

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

357

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 12 • BACKWARD 2 • Q1/5

True or false?

100 less than 612 is 512.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 12 • BACKWARD 2 • Q2/5

362

QUESTION ID

S12-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 12 • BACKWARD 2 • Q2/5

What is 100 more than 285?

- A.** 295
- B.** 385
- C.** 185
- D.** 286

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 12 • BACKWARD 2 • Q3/5

363

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 385

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 12 • BACKWARD 2 • Q3/5

What is 10 less than 503?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 12 • BACKWARD 2 • Q4/5

364

QUESTION ID

S12-B2-Q4

TYPE

Short answer

CORRECT ANSWER

493

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 12 • BACKWARD 2 • Q4/5

True or false?

10 more than 790 is 890.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 12 • BACKWARD 2 • Q5/5

365

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 4.OA.C.5

FORWARD 1

Form ordered pairs from two patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 1 and adds 1. Pattern B starts at 5 and adds 5.

Write the first four ordered pairs (A, B).

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 12 • FORWARD 1 • Q1/5

367

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(1, 5), (2, 10), (3, 15), (4, 20)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 12 • FORWARD 1 • Q1/5

True or false?

Pattern A is 0, 3, 6, 9 and Pattern B is 0, 9, 18, 27.
Each B term is 3 times the matching A term.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 12 • FORWARD 1 • Q2/5

368

QUESTION ID

S12-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 12 • FORWARD 1 • Q2/5

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 8.

What is the B term when the A term is 6?

A. 14

B. 12

C. 48

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 12 • FORWARD 1 • Q3/5

369

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

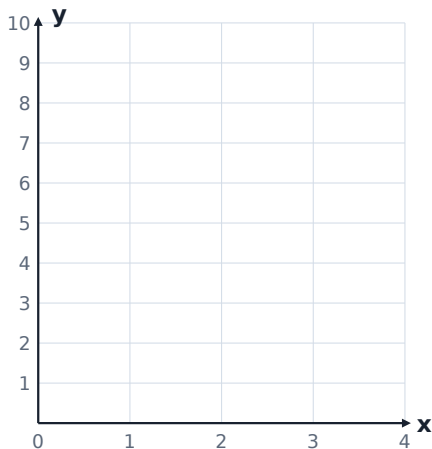
D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 12 • FORWARD 1 • Q3/5

Graph the ordered pairs (0, 0), (1, 3), (2, 6), and (3, 9) from two patterns.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 12 • FORWARD 1 • Q4/5

370

QUESTION ID

S12-F1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (0, 0), (1, 3), (2, 6), (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 12 • FORWARD 1 • Q4/5

True or false?

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10. The 4th pair is (20, 40).

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 12 • FORWARD 1 • Q5/5

371

QUESTION ID

S12-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 12 • FORWARD 1 • Q5/5

SET 12 • 4.OA.C.5

FORWARD 2

Write an equation for a pattern with dependent and independent variables •
Grade 6 • Expressions & Equations • 6.EE.C.9

Figure 1 has 4 tiles, and each figure has 4 more tiles than the one before.

Write an equation for the tiles t in figure n .

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 12 • FORWARD 2 • Q1/5

373

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$t = 4n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 12 • FORWARD 2 • Q1/5

In $t = 4n$, which variable depends on the other?

- A. t
- B. n
- C. 4
- D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 12 • FORWARD 2 • Q2/5

374

QUESTION ID

S12-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 12 • FORWARD 2 • Q2/5

Write an equation for y in terms of x .

x	1	2	3	4
y	6	12	18	24

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 12 • FORWARD 2 • Q3/5

375

QUESTION ID

S12-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$y = 6x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 12 • FORWARD 2 • Q3/5

True or false?

In $c = 8h$, when h increases by 1, c increases by 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 12 • FORWARD 2 • Q4/5

376

QUESTION ID

S12-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 12 • FORWARD 2 • Q4/5

True or false?

The table x: 1, 2, 3 and y: 3, 5, 7 fits $y = 3x$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 12 • FORWARD 2 • Q5/5

377

QUESTION ID

S12-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 12 • FORWARD 2 • Q5/5

Identify features of a pattern that the rule does not state, and explain them

MAIN	Grade 4 main question	4.OA.C.5
BACKWARD 1	Explain patterns in the multiplication and addition tables	3.OA.D.9
BACKWARD 2	Odd and even numbers	2.OA.C.3
FORWARD 1	Explain the relationship between two patterns	5.OA.B.3
FORWARD 2	Evaluate expressions for a given value of the variable	6.EE.A.2.c

SET 13 • 4.OA.C.5

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "add 4," starting at 2: 2, 6, 10, 14, 18.

What do you notice about every term? Explain why it happens.

Feature: _____

Why: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 13 • MAIN • Q1/5

380

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

Every term is even.

GRADING NOTE Why: the pattern starts at an even number and adding 4 (an even number) to an even number always gives an even number. Another true feature with a correct reason (for example, the ones digits repeat 2, 6, 0, 4, 8) is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 13 • MAIN • Q1/5

The rule is "add 5," starting at 3: 3, 8, 13, 18, 23.

Which statement is true?

- A.** Every term is a multiple of 5.
- B.** Every term is even.
- C.** The ones digits alternate between 3 and 8.
- D.** Every term is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 13 • MAIN • Q2/5

381

QUESTION ID

S13-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

**C. The ones digits
alternate between 3
and 8.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 13 • MAIN • Q2/5

True or false?

The rule "add 2," starting at 7, makes only odd numbers.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 13 • MAIN • Q3/5

382

QUESTION ID

S13-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 13 • MAIN • Q3/5

True or false?

The rule "add 6," starting at 1, makes terms that alternate between odd and even.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 13 • MAIN • Q4/5

383

QUESTION ID

S13-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 13 • MAIN • Q4/5

The rule is "multiply by 2," starting at 3: 3, 6, 12, 24, 48.

After the first term, are the terms odd or even? Explain why.

Answer: _____

Why: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 13 • MAIN • Q5/5

384

QUESTION ID

S13-M-Q5

TYPE

Short answer

CORRECT ANSWER

Even

GRADING NOTE Each term after the first is 2 times a whole number, and 2 times any whole number is even.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 13 • MAIN • Q5/5

SET 13 • 4.OA.C.5

BACKWARD 1

Explain patterns in the multiplication and addition tables • Grade 3 •
Operations & Algebraic Thinking • 3.OA.D.9

True or false?

4 times any whole number is even.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 13 • BACKWARD 1 • Q1/5

386

QUESTION ID

S13-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 13 • BACKWARD 1 • Q1/5

Why is every product in the 6s row of the multiplication table even?

- A. The 6s row has 10 products.
- B. 6 is greater than 5.
- C. Every number is even.
- D. 6 is 2×3 , so every product is 2 times a number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 13 • BACKWARD 1 • Q2/5

387

QUESTION ID

S13-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 6 is 2×3 , so every product is 2 times a number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 13 • BACKWARD 1 • Q2/5

The 10s row is 10, 20, 30, 40, ...

What is the same about every product?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 13 • BACKWARD 1 • Q3/5

388

QUESTION ID

S13-B1-Q3

TYPE

Short answer

CORRECT ANSWER

They all end in 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 13 • BACKWARD 1 • Q3/5

True or false?

Every product in the 9s row of the multiplication table is odd.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 13 • BACKWARD 1 • Q4/5

389

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 13 • BACKWARD 1 • Q4/5

Is the sum of two odd numbers odd or even?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 13 • BACKWARD 1 • Q5/5

390

QUESTION ID

S13-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Even

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 4.OA.C.5

BACKWARD 2

Odd and even numbers • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

True or false?

31 is odd.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q1/5

392

QUESTION ID

S13-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Is 46 odd or even?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q2/5

393

QUESTION ID

S13-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Even

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q2/5

Which number is odd?

- A.** 8
- B.** 14
- C.** 30
- D.** 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q3/5

394

QUESTION ID

S13-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q3/5

Write 18 as the sum of two equal addends.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q4/5

395

QUESTION ID

S13-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9 + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q4/5

True or false?

Adding two even numbers can give an odd number.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q5/5

396

QUESTION ID

S13-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 4.OA.C.5

FORWARD 1

Explain the relationship between two patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 6. Pattern B starts at 0 and adds 2.

Explain why each term of A is 3 times the matching term of B.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 13 • FORWARD 1 • Q1/5

398

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Both start at 0 and take the same number of steps, and A adds 6, which is 3×2 , at every step, so A has always added 3 times as much.

GRADING NOTE Must connect the rule amounts ($6 = 3 \times 2$) to the same start of 0 and the same number of steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 13 • FORWARD 1 • Q1/5

True or false?

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 8. Each B term is 2 times the A term because B adds twice as much at every step.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 13 • FORWARD 1 • Q2/5

399

QUESTION ID

S13-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 13 • FORWARD 1 • Q2/5

Pattern A starts at 2 and adds 2. Pattern B starts at 2 and adds 6.

Why is B NOT always 3 times A?

- A.** A has more terms.
- B.** Both rules add even numbers.
- C.** B adds less than A.
- D.** They start at the same number, not at 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 13 • FORWARD 1 • Q3/5

400

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. They start at the same number, not at 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 13 • FORWARD 1 • Q3/5

Pattern A starts at 0 and adds 7. Pattern B starts at 0 and adds 21.

What is the B term when the A term is 35?

Term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 13 • FORWARD 1 • Q4/5

401

QUESTION ID

S13-F1-Q4

TYPE

Short answer

CORRECT ANSWER

105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 13 • FORWARD 1 • Q4/5

True or false?

Two patterns start at 0 and add 5 and 10. Each B term is 5 more than the matching A term.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 13 • FORWARD 1 • Q5/5

402

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 13 • FORWARD 1 • Q5/5

SET 13 • 4.OA.C.5

FORWARD 2

Evaluate expressions for a given value of the variable • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $4n + 2$ when $n = 6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 13 • FORWARD 2 • Q1/5

404

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 13 • FORWARD 2 • Q1/5

True or false?

The expression $2n$ gives an even number for every whole number n .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 13 • FORWARD 2 • Q2/5

405

QUESTION ID

S13-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 13 • FORWARD 2 • Q2/5

The n th term of 5, 9, 13, 17, ... is $4n + 1$.

What is the 10th term?

- A. 40
- B. 45
- C. 41
- D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 13 • FORWARD 2 • Q3/5

406

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 41

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 13 • FORWARD 2 • Q3/5

Evaluate $3n - 1$ when $n = 12$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 13 • FORWARD 2 • Q4/5

407

QUESTION ID

S13-F2-Q4

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 13 • FORWARD 2 • Q4/5

True or false?

When $n = 5$, $2n + 1 = 12$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 13 • FORWARD 2 • Q5/5

408

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 13 • FORWARD 2 • Q5/5

A digit in one place is worth ten times what it is worth in the place to its right

MAIN	Grade 4 main question	4.NBT.A.1
BACKWARD 1	Hundreds, tens, and ones	2.NBT.A.1
BACKWARD 2	Multiply one-digit numbers by multiples of 10	3.NBT.A.3
FORWARD 1	Place value in decimals: 10 times and 1/10 of a digit	5.NBT.A.1
FORWARD 2	Write and evaluate powers of 10 with exponents	6.EE.A.1

SET 14 • 4.NBT.A.1

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.1

True or false?

In 4,440, the 4 in the hundreds place is worth 10 times the 4 in the tens place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 14 • MAIN • Q1/5

411

QUESTION ID

S14-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 14 • MAIN • Q1/5

The 8 in 8,325 is worth how many times the 8 in 825?

- A.** 10 times
- B.** 8 times
- C.** The same
- D.** 100 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 14 • MAIN • Q2/5

412

QUESTION ID

S14-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 10 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 14 • MAIN • Q2/5

What is the value of the 6 in 62,904?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 14 • MAIN • Q3/5

413

QUESTION ID

S14-M-Q3

TYPE

Short answer

CORRECT ANSWER

60,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 14 • MAIN • Q3/5

Write the number that is 10 times as much as 3,070.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 14 • MAIN • Q4/5

414

QUESTION ID

S14-M-Q4

TYPE

Short answer

CORRECT ANSWER

30,700

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 14 • MAIN • Q4/5

True or false?

In 5,555, the 5 in the tens place is worth 10 times the 5 in the hundreds place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 14 • MAIN • Q5/5

415

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 14 • MAIN • Q5/5

SET 14 • 4.NBT.A.1

BACKWARD 1

Hundreds, tens, and ones • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.1

How many tens make 1 hundred?

Tens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 14 • BACKWARD 1 • Q1/5

417

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 14 • BACKWARD 1 • Q1/5

True or false?

In 472, the 7 means 7 tens.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 14 • BACKWARD 1 • Q2/5

418

QUESTION ID

S14-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 14 • BACKWARD 1 • Q2/5

What is the value of the 3 in 360?

- A.** 3,000
- B.** 30
- C.** 3
- D.** 300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 14 • BACKWARD 1 • Q3/5

419

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 14 • BACKWARD 1 • Q3/5

Write the number with 5 hundreds, 0 tens, and 8 ones.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 14 • BACKWARD 1 • Q4/5

420

QUESTION ID

S14-B1-Q4

TYPE

Short answer

CORRECT ANSWER

508

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 14 • BACKWARD 1 • Q4/5

True or false?

In 914, the 1 means 1 hundred.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 14 • BACKWARD 1 • Q5/5

421

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 4.NBT.A.1

BACKWARD 2

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations
in Base Ten • 3.NBT.A.3

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$6 \times 40$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 14 • BACKWARD 2 • Q1/5

423

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

240

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 14 • BACKWARD 2 • Q1/5

True or false?

$$3 \times 70 = 210$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 14 • BACKWARD 2 • Q2/5

424

QUESTION ID

S14-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 14 • BACKWARD 2 • Q2/5

$$9 \times 50 = ?$$

- A.** 45
- B.** 59
- C.** 4,500
- D.** 450

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 14 • BACKWARD 2 • Q3/5

425

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 450

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 14 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$8 \times 30$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 14 • BACKWARD 2 • Q4/5

426

QUESTION ID

S14-B2-Q4

TYPE

Short answer

CORRECT ANSWER

240

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 14 • BACKWARD 2 • Q4/5

426

True or false?

$$5 \times 80 = 4,000$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 14 • BACKWARD 2 • Q5/5

427

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 4.NBT.A.1

FORWARD 1

Place value in decimals: 10 times and $\frac{1}{10}$ of a digit • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1

True or false?

In 3.33, the 3 in the ones place is worth 10 times the 3 in the tenths place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 14 • FORWARD 1 • Q1/5

429

QUESTION ID

S14-F1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 14 • FORWARD 1 • Q1/5

What is $\frac{1}{10}$ of 70?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 14 • FORWARD 1 • Q2/5

430

QUESTION ID

S14-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 14 • FORWARD 1 • Q2/5

In 6.66, the 6 in the hundredths place is worth what part of the 6 in the tenths place?

- A. $\frac{1}{10}$
- B. The same amount
- C. 10 times as much
- D. $\frac{1}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 14 • FORWARD 1 • Q3/5

431

QUESTION ID

S14-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 14 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

What is 10 times 0.4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 14 • FORWARD 1 • Q4/5

432

QUESTION ID

S14-F1-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 14 • FORWARD 1 • Q4/5

432

True or false?

In 2.22, the 2 in the tenths place is worth 10 times the 2 in the ones place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 14 • FORWARD 1 • Q5/5

433

QUESTION ID

S14-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 14 • FORWARD 1 • Q5/5

SET 14 • 4.NBT.A.1

FORWARD 2

Write and evaluate powers of 10 with exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Write $10 \times 10 \times 10$ using an exponent.

Exponent form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 14 • FORWARD 2 • Q1/5

435

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 14 • FORWARD 2 • Q1/5

True or false?

$$10^4 = 10,000$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 14 • FORWARD 2 • Q2/5

436

QUESTION ID

S14-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 14 • FORWARD 2 • Q2/5

Evaluate.

$$10^2$$

- A. 12
- B. 20
- C. 1,000
- D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 14 • FORWARD 2 • Q3/5

437

QUESTION ID

S14-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 14 • FORWARD 2 • Q3/5

Evaluate.

$$4 \times 10^3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 14 • FORWARD 2 • Q4/5

438

QUESTION ID

S14-F2-Q4

TYPE

Short answer

CORRECT ANSWER

4,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 14 • FORWARD 2 • Q4/5

True or false?

10^5 has 4 zeros.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 14 • FORWARD 2 • Q5/5

439

QUESTION ID

S14-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 14 • FORWARD 2 • Q5/5

Use place value and division to explain why one number is ten times another

MAIN	Grade 4 main question	4.NBT.A.1
BACKWARD 1	A hundred is a bundle of ten tens	2.NBT.A.1.a
BACKWARD 2	Multiply one-digit numbers by multiples of 10	3.NBT.A.3
FORWARD 1	Divide whole numbers and decimals by powers of 10	5.NBT.A.2
FORWARD 2	Divide multi-digit decimals	6.NS.B.3

SET 15 • 4.NBT.A.1

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.1

Use place value to find $5,000 \div 500$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 15 • MAIN • Q1/5

442

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 15 • MAIN • Q1/5

True or false?

$900 \div 90 = 10$ because 9 hundreds is 10 times 9 tens.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 15 • MAIN • Q2/5

443

QUESTION ID

S15-M-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 15 • MAIN • Q2/5

Which statement explains why $3,000 \div 300 = 10$?

- A. $3,000 - 300 = 10$
- B. 3 thousands is 10 times 3 hundreds.
- C. 3,000 has one more zero.
- D. $300 \div 3,000 = 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 15 • MAIN • Q3/5

444

QUESTION ID

S15-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3 thousands is 10 times 3 hundreds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 15 • MAIN • Q3/5

4,000 is how many times as much as 400? Explain using place value.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 15 • MAIN • Q4/5

445

QUESTION ID

S15-M-Q4

TYPE

Short answer

CORRECT ANSWER

10 times

GRADING NOTE 4 thousands = 40 hundreds, which is 10 groups of 4 hundreds. Saying only "it has one more zero" is not a place-value explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 15 • MAIN • Q4/5

True or false?

$$60 \div 6 = 100$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 15 • MAIN • Q5/5

446

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 15 • MAIN • Q5/5

SET 15 • 4.NBT.A.1

BACKWARD 1

A hundred is a bundle of ten tens • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a

True or false?

100 can be thought of as a bundle of 10 tens.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a • Set 15 • BACKWARD 1 • Q1/5

448

QUESTION ID

S15-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a • Set 15 • BACKWARD 1 • Q1/5

How many tens are in 300?

Tens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a • Set 15 • BACKWARD 1 • Q2/5

449

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a • Set 15 • BACKWARD 1 • Q2/5

How many hundreds are in 800?

- A.** 8
- B.** 800
- C.** 18
- D.** 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a • Set 15 • BACKWARD 1 • Q3/5

450

QUESTION ID

S15-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a • Set 15 • BACKWARD 1 • Q3/5

How many tens are in 50?

Tens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a • Set 15 • BACKWARD 1 • Q4/5

451

QUESTION ID

S15-B1-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a • Set 15 • BACKWARD 1 • Q4/5

True or false?

400 is 4 tens.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a • Set 15 • BACKWARD 1 • Q5/5

452

QUESTION ID

S15-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 4.NBT.A.1

BACKWARD 2

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$4 \times 90$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 15 • BACKWARD 2 • Q1/5

454

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

360

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 15 • BACKWARD 2 • Q1/5

True or false?

$$7 \times 60 = 420$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 15 • BACKWARD 2 • Q2/5

455

QUESTION ID

S15-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 15 • BACKWARD 2 • Q2/5

$$2 \times 80 = ?$$

- A.** 160
- B.** 82
- C.** 16
- D.** 1,600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 15 • BACKWARD 2 • Q3/5

456

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 160

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 15 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$9 \times 20$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 15 • BACKWARD 2 • Q4/5

457

QUESTION ID

S15-B2-Q4

TYPE

Short answer

CORRECT ANSWER

180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 15 • BACKWARD 2 • Q4/5

457

True or false?

$$6 \times 50 = 3,000$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 15 • BACKWARD 2 • Q5/5

458

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 4.NBT.A.1

FORWARD 1

Divide whole numbers and decimals by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$4,500 \div 100$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 15 • FORWARD 1 • Q1/5

460

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 15 • FORWARD 1 • Q1/5

True or false?

$$7,000 \div 10^3 = 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 15 • FORWARD 1 • Q2/5

461

QUESTION ID

S15-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 15 • FORWARD 1 • Q2/5

Divide.

$$36 \div 10$$

- A.** 360
- B.** 3.6
- C.** 36
- D.** 0.36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 15 • FORWARD 1 • Q3/5

462

QUESTION ID

S15-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 15 • FORWARD 1 • Q3/5

Divide.

$$820 \div 10^2$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 15 • FORWARD 1 • Q4/5

463

QUESTION ID

S15-F1-Q4

TYPE

Short answer

CORRECT ANSWER

8.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 15 • FORWARD 1 • Q4/5

True or false?

$$50 \div 10 = 500$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 15 • FORWARD 1 • Q5/5

464

QUESTION ID

S15-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 4.NBT.A.1

FORWARD 2

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$4.5 \div 0.45$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 2 • Q1/5

466

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 2 • Q1/5

466

True or false?

$$0.8 \div 0.08 = 10$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 2 • Q2/5

467

QUESTION ID

S15-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 2 • Q2/5

Divide.

$$6 \div 0.6$$

- A.** 10
- B.** 0.1
- C.** 1
- D.** 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 2 • Q3/5

468

QUESTION ID

S15-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$2.4 \div 0.3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 2 • Q4/5

469

QUESTION ID

S15-F2-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 2 • Q4/5

469

True or false?

$$1.2 \div 0.12 = 100$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 2 • Q5/5

470

QUESTION ID

S15-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 2 • Q5/5

Read and write multi-digit whole numbers using numerals and number names

MAIN	Grade 4 main question	4.NBT.A.2
BACKWARD 1	Read and write numbers to 1000	2.NBT.A.3
BACKWARD 2	Hundreds, tens, and ones	2.NBT.A.1
FORWARD 1	Read and write decimals to thousandths	5.NBT.A.3.a
FORWARD 2	Use positive and negative numbers to describe quantities — near...	6.NS.C.5

SET 16 • 4.NBT.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.2

Write the number in standard form.

forty-six thousand, two hundred eight

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 16 • MAIN • Q1/5

473

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

46,208

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 16 • MAIN • Q1/5

Write 309,051 in words.

Words: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 16 • MAIN • Q2/5

474

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

**three hundred nine
thousand, fifty-one**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 16 • MAIN • Q2/5

Which is 72,016 in words?

- A.** seventy-two thousand, sixteen
- B.** seven thousand, two hundred sixteen
- C.** seventy-two thousand, one hundred six
- D.** seventy-two thousand, one hundred sixty

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 16 • MAIN • Q3/5

475

QUESTION ID

S16-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. seventy-two thousand, sixteen

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 16 • MAIN • Q3/5

True or false?

Six hundred thousand, forty is written 600,400.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 16 • MAIN • Q4/5

476

QUESTION ID

S16-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 16 • MAIN • Q4/5

Write the number in standard form.

ninety thousand, nine hundred

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 16 • MAIN • Q5/5

477

QUESTION ID

S16-M-Q5

TYPE

Short answer

CORRECT ANSWER

90,900

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 16 • MAIN • Q5/5

SET 16 • 4.NBT.A.2

BACKWARD 1

Read and write numbers to 1000 • Grade 2 • Number & Operations in Base Ten
• 2.NBT.A.3

Write the number in standard form.

four hundred seventeen

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 16 • BACKWARD 1 • Q1/5

479

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

417

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 16 • BACKWARD 1 • Q1/5

True or false?

Six hundred five is written 650.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 16 • BACKWARD 1 • Q2/5

480

QUESTION ID

S16-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 16 • BACKWARD 1 • Q2/5

Which is 238 in words?

- A.** two hundred thirty-eight
- B.** three hundred twenty-eight
- C.** two hundred eighty-three
- D.** twenty-three eight

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 16 • BACKWARD 1 • Q3/5

481

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

**A. two hundred
thirty-eight**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 16 • BACKWARD 1 • Q3/5

Write 702 in words.

Words: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 16 • BACKWARD 1 • Q4/5

482

QUESTION ID

S16-B1-Q4

TYPE

Short answer

CORRECT ANSWER

seven hundred two

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 16 • BACKWARD 1 • Q4/5

True or false?

Nine hundred ninety is written 990.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 16 • BACKWARD 1 • Q5/5

483

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 4.NBT.A.2

BACKWARD 2

Hundreds, tens, and ones • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.1

What digit is in the tens place of 586?

Digit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 16 • BACKWARD 2 • Q1/5

485

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 16 • BACKWARD 2 • Q1/5

True or false?

In 249, the 2 is worth 200.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 16 • BACKWARD 2 • Q2/5

486

QUESTION ID

S16-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 16 • BACKWARD 2 • Q2/5

Which number has 7 hundreds, 3 tens, and 0 ones?

- A.** 730
- B.** 703
- C.** 73
- D.** 370

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 16 • BACKWARD 2 • Q3/5

487

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 730

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 16 • BACKWARD 2 • Q3/5

How many hundreds are in 600?

Hundreds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 16 • BACKWARD 2 • Q4/5

488

QUESTION ID

S16-B2-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 16 • BACKWARD 2 • Q4/5

True or false?

In 381, the 8 is worth 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 16 • BACKWARD 2 • Q5/5

489

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 4.NBT.A.2

FORWARD 1

Read and write decimals to thousandths • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a

Write the number in standard form.

four and twenty-five thousandths

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • FORWARD 1 • Q1/5

491

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4.025

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • FORWARD 1 • Q1/5

True or false?

0.6 is read "six tenths."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • FORWARD 1 • Q2/5

492

QUESTION ID

S16-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • FORWARD 1 • Q2/5

Which is 3.48 in words?

- A.** thirty-four and eight tenths
- B.** three hundred forty-eight
- C.** three and forty-eight tenths
- D.** three and forty-eight hundredths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • FORWARD 1 • Q3/5

493

QUESTION ID

S16-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

**D. three and
forty-eight
hundredths**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • FORWARD 1 • Q3/5

Write 12.307 in words.

Words: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • FORWARD 1 • Q4/5

494

QUESTION ID

S16-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**twelve and three
hundred seven
thousandths**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • FORWARD 1 • Q4/5

True or false?

Five and nine hundredths is written 5.9.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • FORWARD 1 • Q5/5

495

QUESTION ID

S16-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • FORWARD 1 • Q5/5

FORWARD 2

Use positive and negative numbers to describe quantities • Grade 6 • The Number System • 6.NS.C.5

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

A diver is 30 feet below sea level.

Write the diver's position as an integer.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 16 • FORWARD 2 • Q1/5 • NEAREST RELATED

497

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 16 • FORWARD 2 • Q1/5 • NEAREST RELATED

True or false?

A gain of 8 yards can be written as +8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 16 • FORWARD 2 • Q2/5 • NEAREST RELATED

498

QUESTION ID

S16-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 16 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which integer shows a temperature of 12 degrees below zero?

- A.** -12
- B.** 12
- C.** 0
- D.** -21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 16 • FORWARD 2 • Q3/5 • NEAREST RELATED

499

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 16 • FORWARD 2 • Q3/5 • NEAREST RELATED

A deposit of \$45 is written as +45.

How is a withdrawal of \$45 written?

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 16 • FORWARD 2 • Q4/5 • NEAREST RELATED

500

QUESTION ID

S16-F2-Q4

TYPE

Short answer

CORRECT ANSWER

−45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 16 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

–5 means 5 more than zero.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 16 • FORWARD 2 • Q5/5 • NEAREST RELATED

501

QUESTION ID

S16-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 16 • FORWARD 2 • Q5/5 • NEAREST RELATED

Write multi-digit whole numbers in expanded form

MAIN	Grade 4 main question	4.NBT.A.2
BACKWARD 1	Expanded form for numbers to 1000	2.NBT.A.3
BACKWARD 2	Two-digit numbers as tens and ones	1.NBT.B.2
FORWARD 1	Write decimals in expanded form	5.NBT.A.3.a
FORWARD 2	Write numbers using powers of 10 with exponents	6.EE.A.1

SET 17 • 4.NBT.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.2

Write 7,345 in expanded form.

Expanded form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 17 • MAIN • Q1/5

504

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

7,000 + 300 + 40 + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 17 • MAIN • Q1/5

Write the number.

$$50,000 + 2,000 + 80 + 6$$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 17 • MAIN • Q2/5

505

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

52,086

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 17 • MAIN • Q2/5

Which is the expanded form of 40,209?

- A. $40,000 + 2,000 + 9$
- B. $40,000 + 20 + 9$
- C. $40,000 + 200 + 9$
- D. $4,000 + 200 + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 17 • MAIN • Q3/5

506

QUESTION ID

S17-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $40,000 + 200 + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 17 • MAIN • Q3/5

True or false?

$$600,000 + 3,000 + 70 = 630,070$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 17 • MAIN • Q4/5

507

QUESTION ID

S17-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 17 • MAIN • Q4/5

Write 18,504 in expanded form.

Expanded form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 17 • MAIN • Q5/5

508

QUESTION ID

S17-M-Q5

TYPE

Short answer

CORRECT ANSWER

**10,000 + 8,000
+ 500 + 4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 17 • MAIN • Q5/5

SET 17 • 4.NBT.A.2

BACKWARD 1

Expanded form for numbers to 1000 • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3

Write 846 in expanded form.

Expanded form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 17 • BACKWARD 1 • Q1/5

510

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

800 + 40 + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 17 • BACKWARD 1 • Q1/5

True or false?

$$300 + 9 = 390$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 17 • BACKWARD 1 • Q2/5

511

QUESTION ID

S17-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 17 • BACKWARD 1 • Q2/5

Which number is $500 + 20 + 7$?

- A.** 527
- B.** 572
- C.** 257
- D.** 5,207

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 17 • BACKWARD 1 • Q3/5

512

QUESTION ID

S17-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 527

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 17 • BACKWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Write the number.

$$900 + 60$$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 17 • BACKWARD 1 • Q4/5

513

QUESTION ID

S17-B1-Q4

TYPE

Short answer

CORRECT ANSWER

960

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 17 • BACKWARD 1 • Q4/5

513

True or false?

$$200 + 40 + 1 = 241$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 17 • BACKWARD 1 • Q5/5

514

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 4.NBT.A.2

BACKWARD 2

Two-digit numbers as tens and ones • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2

How many tens and ones are in 47?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2 • Set 17 • BACKWARD 2 • Q1/5

516

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4 tens and 7 ones

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2 • Set 17 • BACKWARD 2 • Q1/5

True or false?

63 is 6 tens and 3 ones.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2 • Set 17 • BACKWARD 2 • Q2/5

517

QUESTION ID

S17-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2 • Set 17 • BACKWARD 2 • Q2/5

Which number is 8 tens and 2 ones?

- A.** 28
- B.** 10
- C.** 82
- D.** 802

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2 • Set 17 • BACKWARD 2 • Q3/5

518

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 82

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2 • Set 17 • BACKWARD 2 • Q3/5

Write the number that is 5 tens and 0 ones.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2 • Set 17 • BACKWARD 2 • Q4/5

519

QUESTION ID

S17-B2-Q4

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2 • Set 17 • BACKWARD 2 • Q4/5

True or false?

91 is 1 ten and 9 ones.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2 • Set 17 • BACKWARD 2 • Q5/5

520

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 4.NBT.A.2

FORWARD 1

Write decimals in expanded form • Grade 5 • Number & Operations in Base Ten
• 5.NBT.A.3.a

Write 3.46 in expanded form.

Expanded form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • FORWARD 1 • Q1/5

522

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$$3 \times 1 + 4 \times (1/10) + 6 \times (1/100)$$

GRADING NOTE 3 + 0.4 + 0.06 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • FORWARD 1 • Q1/5

True or false?

$$2 + 0.5 + 0.03 = 2.53$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • FORWARD 1 • Q2/5

523

QUESTION ID

S17-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • FORWARD 1 • Q2/5

Which number is $7 \times 1 + 2 \times \frac{1}{10} + 9 \times \frac{1}{1000}$?

- A. 7.029
- B. 7.209
- C. 72.9
- D. 7.29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • FORWARD 1 • Q3/5

524

QUESTION ID

S17-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7.209

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • FORWARD 1 • Q3/5

Write the number.

$$5 \times 10 + 1 \times 1 + 8 \times \frac{1}{100}$$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • FORWARD 1 • Q4/5

525

QUESTION ID

S17-F1-Q4

TYPE

Short answer

CORRECT ANSWER

51.08

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • FORWARD 1 • Q4/5

True or false?

$$4 \times \frac{1}{10} + 1 \times \frac{1}{100} = 0.041$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • FORWARD 1 • Q5/5

526

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • FORWARD 1 • Q5/5

SET 17 • 4.NBT.A.2

FORWARD 2

Write numbers using powers of 10 with exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Write $6,000 + 400 + 70 + 2$ using powers of 10.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • FORWARD 2 • Q1/5

528

QUESTION ID

S17-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$$6 \times 10^3 + 4 \times 10^2 \\ + 7 \times 10^1 + 2$$

GRADING NOTE Writing the last term as 2×10^0 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • FORWARD 2 • Q1/5

True or false?

$$3 \times 10^4 = 30,000$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • FORWARD 2 • Q2/5

529

QUESTION ID

S17-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • FORWARD 2 • Q2/5

Which number equals $5 \times 10^3 + 8 \times 10$?

- A. 580
- B. 5,008
- C. 5,800
- D. 5,080

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • FORWARD 2 • Q3/5

530

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5,080

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • FORWARD 2 • Q3/5

Evaluate.

$$2 \times 10^3 + 9 \times 10^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • FORWARD 2 • Q4/5

531

QUESTION ID

S17-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2,900

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • FORWARD 2 • Q4/5

True or false?

$$7 \times 10^2 = 70$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • FORWARD 2 • Q5/5

532

QUESTION ID

S17-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • FORWARD 2 • Q5/5

Compare multi-digit whole numbers using $>$, $=$, and $<$

MAIN	Grade 4 main question	4.NBT.A.2
BACKWARD 1	Compare three-digit numbers	2.NBT.A.4
BACKWARD 2	Compare two-digit numbers	1.NBT.B.3
FORWARD 1	Compare decimals to thousandths	5.NBT.A.3.b
FORWARD 2	Order rational numbers using their positions on a number line	6.NS.C.7.a

SET 18 • 4.NBT.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.2

ANSWER KEY

Reduced copy of the question page, with its answer

Write $>$, $=$, or $<$.

45,621 ____ 45,612

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 18 • MAIN • Q1/5

535

QUESTION ID

S18-M-Q1

TYPE

Short answer

CORRECT ANSWER

 $>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 18 • MAIN • Q1/5

535

Which comparison is true?

- A. $80,099 > 80,100$
- B. $80,099 = 80,100$
- C. $80,100 < 80,099$
- D. $80,099 < 80,100$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 18 • MAIN • Q2/5

536

QUESTION ID

S18-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $80,099 < 80,100$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 18 • MAIN • Q2/5

True or false?

305,000 < 350,000

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 18 • MAIN • Q3/5

537

QUESTION ID

S18-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 18 • MAIN • Q3/5

Order from least to greatest.

27,450 24,750 27,045

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 18 • MAIN • Q4/5

538

QUESTION ID

S18-M-Q4

TYPE

Short answer

CORRECT ANSWER

**24,750, 27,045,
27,450**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 18 • MAIN • Q4/5

True or false?

9,998 > 10,001

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 18 • MAIN • Q5/5

539

QUESTION ID

S18-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 18 • MAIN • Q5/5

SET 18 • 4.NBT.A.2

BACKWARD 1

Compare three-digit numbers • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

Write $>$, $=$, or $<$.

482 ____ 428

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 18 • BACKWARD 1 • Q1/5

541

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 18 • BACKWARD 1 • Q1/5

True or false?

$309 < 390$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 18 • BACKWARD 1 • Q2/5

542

QUESTION ID

S18-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 18 • BACKWARD 1 • Q2/5

Which number is greatest?

- A.** 170
- B.** 699
- C.** 701
- D.** 710

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 18 • BACKWARD 1 • Q3/5

543

QUESTION ID

S18-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 710

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 18 • BACKWARD 1 • Q3/5

Write $>$, $=$, or $<$.

555 ____ 555

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 18 • BACKWARD 1 • Q4/5

544

QUESTION ID

S18-B1-Q4

TYPE

Short answer

CORRECT ANSWER

=

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 18 • BACKWARD 1 • Q4/5

True or false?

640 > 664

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 18 • BACKWARD 1 • Q5/5

545

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 4.NBT.A.2

BACKWARD 2

Compare two-digit numbers • Grade 1 • Number & Operations in Base Ten •
1.NBT.B.3

Write $>$, $=$, or $<$.

67 ____ 76

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 18 • BACKWARD 2 • Q1/5

547

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 18 • BACKWARD 2 • Q1/5

True or false?

58 > 49

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 18 • BACKWARD 2 • Q2/5

548

QUESTION ID

S18-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 18 • BACKWARD 2 • Q2/5

Which number is less than 35?

- A.** 35
- B.** 53
- C.** 29
- D.** 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 18 • BACKWARD 2 • Q3/5

549

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 18 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Write $>$, $=$, or $<$.

90 ____ 19

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 18 • BACKWARD 2 • Q4/5

550

QUESTION ID

S18-B2-Q4

TYPE

Short answer

CORRECT ANSWER

 $>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 18 • BACKWARD 2 • Q4/5

550

True or false?

$$42 < 24$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 18 • BACKWARD 2 • Q5/5

551

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 4.NBT.A.2

FORWARD 1

Compare decimals to thousandths • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b

ANSWER KEY

Reduced copy of the question page, with its answer

Write $>$, $=$, or $<$.

0.45 ____ 0.405

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 18 • FORWARD 1 • Q1/5

553

QUESTION ID

S18-F1-Q1

TYPE

Short answer

CORRECT ANSWER

 $>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 18 • FORWARD 1 • Q1/5

553

True or false?

$3.09 < 3.1$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 18 • FORWARD 1 • Q2/5

554

QUESTION ID

S18-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 18 • FORWARD 1 • Q2/5

Which decimal is greatest?

- A.** 2.5
- B.** 2.09
- C.** 2.405
- D.** 2.48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 18 • FORWARD 1 • Q3/5

555

QUESTION ID

S18-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 18 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Write $>$, $=$, or $<$.

6.70 ____ 6.7

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 18 • FORWARD 1 • Q4/5

556

QUESTION ID

S18-F1-Q4

TYPE

Short answer

CORRECT ANSWER

=

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 18 • FORWARD 1 • Q4/5

556

True or false?

$0.125 > 0.13$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 18 • FORWARD 1 • Q5/5

557

QUESTION ID

S18-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 18 • FORWARD 1 • Q5/5

SET 18 • 4.NBT.A.2

FORWARD 2

Order rational numbers using their positions on a number line • Grade 6 • The Number System • 6.NS.C.7.a

Write $>$, $=$, or $<$.

-8 ____ -3

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 18 • FORWARD 2 • Q1/5

559

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 18 • FORWARD 2 • Q1/5

True or false?

$$-2 > -10$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 18 • FORWARD 2 • Q2/5

560

QUESTION ID

S18-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 18 • FORWARD 2 • Q2/5

Which number is least?

- A. 0
- B. -5
- C. 5
- D. -15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 18 • FORWARD 2 • Q3/5

561

QUESTION ID

S18-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. -15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 18 • FORWARD 2 • Q3/5

Order from least to greatest.

4, -6, 0, -1

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 18 • FORWARD 2 • Q4/5

562

QUESTION ID

S18-F2-Q4

TYPE

Short answer

CORRECT ANSWER

-6, -1, 0, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 18 • FORWARD 2 • Q4/5

True or false?

$$-7 > 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 18 • FORWARD 2 • Q5/5

563

QUESTION ID

S18-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 18 • FORWARD 2 • Q5/5

Round multi-digit whole numbers to any place

MAIN	Grade 4 main question	4.NBT.A.3
BACKWARD 1	Round to the nearest 10 or 100	3.NBT.A.1
BACKWARD 2	Skip-count by 5s, 10s, and 100s	2.NBT.A.2
FORWARD 1	Round decimals to any place	5.NBT.A.4
FORWARD 2	Compute with decimals and round the result — nearest related (n...	6.NS.B.3

SET 19 • 4.NBT.A.3

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.3

Round 46,829 to the nearest thousand.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 19 • MAIN • Q1/5

566

QUESTION ID

S19-M-Q1

TYPE

Short answer

CORRECT ANSWER

47,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 19 • MAIN • Q1/5

Round 352,618 to the nearest ten thousand.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 19 • MAIN • Q2/5

567

QUESTION ID

S19-M-Q2

TYPE

Short answer

CORRECT ANSWER

350,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 19 • MAIN • Q2/5

Round 7,950 to the nearest hundred.

- A. 7,900
- B. 7,960
- C. 7,000
- D. 8,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 19 • MAIN • Q3/5

568

QUESTION ID

S19-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 19 • MAIN • Q3/5

True or false?

128,476 rounded to the nearest hundred thousand is 200,000.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 19 • MAIN • Q4/5

569

QUESTION ID

S19-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 19 • MAIN • Q4/5

A stadium has 63,548 seats.

Round the number of seats to the nearest thousand.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 19 • MAIN • Q5/5

570

QUESTION ID

S19-M-Q5

TYPE

Short answer

CORRECT ANSWER

64,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 19 • MAIN • Q5/5

SET 19 • 4.NBT.A.3

BACKWARD 1

Round to the nearest 10 or 100 • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.1

Round 84 to the nearest ten.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 19 • BACKWARD 1 • Q1/5

572

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 19 • BACKWARD 1 • Q1/5

True or false?

652 rounded to the nearest hundred is 700.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 19 • BACKWARD 1 • Q2/5

573

QUESTION ID

S19-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 19 • BACKWARD 1 • Q2/5

Round 345 to the nearest ten.

- A.** 400
- B.** 350
- C.** 300
- D.** 340

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 19 • BACKWARD 1 • Q3/5

574

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 350

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 19 • BACKWARD 1 • Q3/5

Round 961 to the nearest hundred.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 19 • BACKWARD 1 • Q4/5

575

QUESTION ID

S19-B1-Q4

TYPE

Short answer

CORRECT ANSWER

1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 19 • BACKWARD 1 • Q4/5

True or false?

26 rounded to the nearest ten is 20.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 19 • BACKWARD 1 • Q5/5

576

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • 4.NBT.A.3

BACKWARD 2

Skip-count by 5s, 10s, and 100s • Grade 2 • Number & Operations in Base Ten
• 2.NBT.A.2

Skip-count by 100s.

300, 400, 500, ____

Next: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 19 • BACKWARD 2 • Q1/5

578

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 19 • BACKWARD 2 • Q1/5

True or false?

Counting by 10s from 470 gives 480, 490, 500.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 19 • BACKWARD 2 • Q2/5

579

QUESTION ID

S19-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 19 • BACKWARD 2 • Q2/5

What comes next when counting by 5s?

85, 90, 95, ____

- A.** 96
- B.** 105
- C.** 100
- D.** 99

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 19 • BACKWARD 2 • Q3/5

580

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 19 • BACKWARD 2 • Q3/5

Skip-count by 10s.

620, 630, 640, ____

Next: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 19 • BACKWARD 2 • Q4/5

581

QUESTION ID

S19-B2-Q4

TYPE

Short answer

CORRECT ANSWER

650

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 19 • BACKWARD 2 • Q4/5

True or false?

Counting by 100s from 700 gives 800, 900, 990.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 19 • BACKWARD 2 • Q5/5

582

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 4.NBT.A.3

FORWARD 1

Round decimals to any place • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 3.468 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 1 • Q1/5

584

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 1 • Q1/5

Round 14.729 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 1 • Q2/5

585

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 1 • Q2/5

Round 0.638 to the nearest hundredth.

- A. 0.6
- B. 0.64
- C. 0.7
- D. 0.63

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 1 • Q3/5

586

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 1 • Q3/5

True or false?

5.95 rounded to the nearest tenth is 6.0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 1 • Q4/5

587

QUESTION ID

S19-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 1 • Q4/5

True or false?

8.249 rounded to the nearest tenth is 8.3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 1 • Q5/5

588

QUESTION ID

S19-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 1 • Q5/5

FORWARD 2

Compute with decimals and round the result • Grade 6 • The Number System •
6.NS.B.3

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

Divide. Round the quotient to the nearest tenth.

$$10 \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 2 • Q1/5 • NEAREST RELATED

590

QUESTION ID

S19-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 2 • Q1/5 • NEAREST RELATED

True or false?

$$7.25 \times 4 = 29$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 2 • Q2/5 • NEAREST RELATED

591

QUESTION ID

S19-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 2 • Q2/5 • NEAREST RELATED

Divide. Round the quotient to the nearest hundredth.

$$2 \div 7$$

- A. 0.27
- B. 0.29
- C. 0.28
- D. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 2 • Q3/5 • NEAREST RELATED

592

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 2 • Q3/5 • NEAREST RELATED

Multiply. Round the product to the nearest whole number.

$$3.6 \times 2.4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 2 • Q4/5 • NEAREST RELATED

593

QUESTION ID

S19-F2-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

15 ÷ 4, rounded to the nearest whole number, is 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 2 • Q5/5 • NEAREST RELATED

594

QUESTION ID

S19-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 2 • Q5/5 • NEAREST RELATED

Add multi-digit whole numbers using the standard algorithm

MAIN	Grade 4 main question	4.NBT.B.4
BACKWARD 1	Add within 1000	3.NBT.A.2
BACKWARD 2	Add by place value, making a ten or a hundred	2.NBT.B.7
FORWARD 1	Add decimals to hundredths	5.NBT.B.7
FORWARD 2	Add multi-digit decimals with the standard algorithm	6.NS.B.3

SET 20 • 4.NBT.B.4

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Use the standard algorithm to add. Show your work.

$$4,587 + 2,946$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 20 • MAIN • Q1/5

597

QUESTION ID

S20-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

7,533

GRADING NOTE REQUIRED WORK: The standard algorithm: digits lined up by place value and added from the ones place to the left, with each regrouping shown. Another method earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 20 • MAIN • Q1/5

Use the standard algorithm to add. Show your work.

$$38,209 + 15,894$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 20 • MAIN • Q2/5

598

QUESTION ID

S20-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

54,103

GRADING NOTE REQUIRED WORK: The standard algorithm: digits lined up by place value and added from the ones place to the left, with each regrouping shown. Another method earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 20 • MAIN • Q2/5

Add.

$$6,078 + 3,945$$

- A. 9,023
- B. 9,923
- C. 10,013
- D. 10,023

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 20 • MAIN • Q3/5

599

QUESTION ID

S20-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10,023

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 20 • MAIN • Q3/5

A library had 12,486 books. It got 3,758 more.

Use the standard algorithm to find how many books it has now. Show your work.

Books: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 20 • MAIN • Q4/5

600

QUESTION ID

S20-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

16,244

GRADING NOTE REQUIRED WORK: The standard algorithm: digits lined up by place value and added from the ones place to the left, with each regrouping shown. Another method earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 20 • MAIN • Q4/5

True or false?

In the standard algorithm for $3,685 + 1,476$, you regroup 11 ones as 1 ten and 1 one.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 20 • MAIN • Q5/5

601

QUESTION ID

S20-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 20 • MAIN • Q5/5

SET 20 • 4.NBT.B.4

BACKWARD 1

Add within 1000 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2

Add.

$$356 + 478$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 20 • BACKWARD 1 • Q1/5

603

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

834

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 20 • BACKWARD 1 • Q1/5

True or false?

$$609 + 295 = 904$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 20 • BACKWARD 1 • Q2/5

604

QUESTION ID

S20-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 20 • BACKWARD 1 • Q2/5

$$147 + 268 = ?$$

- A. 415
- B. 315
- C. 405
- D. 425

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 20 • BACKWARD 1 • Q3/5

605

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 415

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 20 • BACKWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$523 + 389$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 20 • BACKWARD 1 • Q4/5

606

QUESTION ID

S20-B1-Q4

TYPE

Short answer

CORRECT ANSWER

912

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 20 • BACKWARD 1 • Q4/5

True or false?

$$480 + 360 = 740$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 20 • BACKWARD 1 • Q5/5

607

QUESTION ID

S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • 4.NBT.B.4

BACKWARD 2

Add by place value, making a ten or a hundred • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7

True or false?

7 ones and 5 ones make 1 ten and 2 ones.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 20 • BACKWARD 2 • Q1/5

609

QUESTION ID

S20-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 20 • BACKWARD 2 • Q1/5

Rename 16 tens as hundreds and tens.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 20 • BACKWARD 2 • Q2/5

610

QUESTION ID

S20-B2-Q2

TYPE

Short answer

CORRECT ANSWER

**1 hundred and 6
tens**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 20 • BACKWARD 2 • Q2/5

Add the hundreds, the tens, and the ones.

$$248 + 135$$

- A. 483
- B. 373
- C. 383
- D. 313

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 20 • BACKWARD 2 • Q3/5

611

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 383

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 20 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$200 + 70 + 4 + 300 + 20 + 9$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 20 • BACKWARD 2 • Q4/5

612

QUESTION ID

S20-B2-Q4

TYPE

Short answer

CORRECT ANSWER

603

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 20 • BACKWARD 2 • Q4/5

612

True or false?

12 ones is the same as 2 tens.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 20 • BACKWARD 2 • Q5/5

613

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 4.NBT.B.4

FORWARD 1

Add decimals to hundredths • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$35.68 + 7.45$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • FORWARD 1 • Q1/5

615

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

43.13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • FORWARD 1 • Q1/5

615

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$4.9 + 12.36$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • FORWARD 1 • Q2/5

616

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

17.26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • FORWARD 1 • Q2/5

616

Add.

$$0.75 + 0.48$$

- A.** 1.13
- B.** 12.3
- C.** 1.23
- D.** 0.123

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • FORWARD 1 • Q3/5

617

QUESTION ID

S20-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1.23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • FORWARD 1 • Q3/5

True or false?

$$6.07 + 3.9 = 9.97$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • FORWARD 1 • Q4/5

618

QUESTION ID

S20-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • FORWARD 1 • Q4/5

True or false?

$$2.5 + 1.75 = 3.80$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • FORWARD 1 • Q5/5

619

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • FORWARD 1 • Q5/5

SET 20 • 4.NBT.B.4

FORWARD 2

Add multi-digit decimals with the standard algorithm • Grade 6 • The Number System • 6.NS.B.3

Add.

$$128.406 + 75.89$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 2 • Q1/5

621

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

204.296

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$9.075 + 0.98$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 2 • Q2/5

622

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10.055

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 2 • Q2/5

622

Add.

$$46.3 + 8.759$$

- A.** 55.059
- B.** 131.89
- C.** 55.159
- D.** 54.059

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 2 • Q3/5

623

QUESTION ID

S20-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 55.059

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 2 • Q3/5

True or false?

$$0.625 + 0.375 = 1$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 2 • Q4/5

624

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 2 • Q4/5

True or false?

$$12.48 + 3.6 = 12.84$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 2 • Q5/5

625

QUESTION ID

S20-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 2 • Q5/5

Subtract multi-digit whole numbers using the standard algorithm

MAIN	Grade 4 main question	4.NBT.B.4
BACKWARD 1	Subtract within 1000	3.NBT.A.2
BACKWARD 2	Subtract by breaking apart a ten or a hundred	2.NBT.B.7
FORWARD 1	Subtract decimals to hundredths	5.NBT.B.7
FORWARD 2	Subtract multi-digit decimals with the standard algorithm	6.NS.B.3

SET 21 • 4.NBT.B.4

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Use the standard algorithm to subtract. Show your work.

$$8,253 - 4,678$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 21 • MAIN • Q1/5

628

QUESTION ID

S21-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

3,575

GRADING NOTE REQUIRED WORK: The standard algorithm: digits lined up by place value and subtracted from the ones place to the left, with each regrouping (renaming) shown. Another method earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 21 • MAIN • Q1/5

Use the standard algorithm to subtract. Show your work.

$$60,000 - 27,438$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 21 • MAIN • Q2/5

629

QUESTION ID

S21-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

32,562

GRADING NOTE REQUIRED WORK: The standard algorithm: digits lined up by place value and subtracted from the ones place to the left, with each regrouping (renaming) shown. Another method earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 21 • MAIN • Q2/5

Subtract.

$$7,012 - 3,456$$

- A. 4,656
- B. 3,556
- C. 4,444
- D. 3,566

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 21 • MAIN • Q3/5

630

QUESTION ID

S21-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3,556

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 21 • MAIN • Q3/5

A school raised \$15,250. It spent \$8,795.

Use the standard algorithm to find how much money is left. Show your work.

Money left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 21 • MAIN • Q4/5

631

QUESTION ID

S21-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

\$6,455

GRADING NOTE REQUIRED WORK: The standard algorithm: digits lined up by place value and subtracted from the ones place to the left, with each regrouping (renaming) shown. Another method earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 21 • MAIN • Q4/5

True or false?

To find $5,042 - 1,317$ with the standard algorithm, you rename 1 ten as 10 ones.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 21 • MAIN • Q5/5

632

QUESTION ID

S21-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 21 • MAIN • Q5/5

SET 21 • 4.NBT.B.4

BACKWARD 1

Subtract within 1000 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$732 - 485$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 21 • BACKWARD 1 • Q1/5

634

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

247

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 21 • BACKWARD 1 • Q1/5

634

True or false?

$$600 - 278 = 322$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 21 • BACKWARD 1 • Q2/5

635

QUESTION ID

S21-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 21 • BACKWARD 1 • Q2/5

$$514 - 296 = ?$$

- A. 222
- B. 228
- C. 218
- D. 318

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 21 • BACKWARD 1 • Q3/5

636

QUESTION ID

S21-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 218

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 21 • BACKWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$905 - 467$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 21 • BACKWARD 1 • Q4/5

637

QUESTION ID

S21-B1-Q4

TYPE

Short answer

CORRECT ANSWER

438

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 21 • BACKWARD 1 • Q4/5

637

True or false?

$$840 - 390 = 550$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 21 • BACKWARD 1 • Q5/5

638

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 4.NBT.B.4

BACKWARD 2

Subtract by breaking apart a ten or a hundred • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7

True or false?

To find $52 - 8$, you can rename 1 ten as 10 ones.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 21 • BACKWARD 2 • Q1/5

640

QUESTION ID

S21-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 21 • BACKWARD 2 • Q1/5

Rename 4 hundreds 2 tens as 3 hundreds and how many tens?

Tens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 21 • BACKWARD 2 • Q2/5

641

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 21 • BACKWARD 2 • Q2/5

Which shows 300 renamed so you can subtract 126?

- A.** 2 hundreds 10 ones
- B.** 3 hundreds 10 tens
- C.** 30 tens 1 one
- D.** 2 hundreds 9 tens 10 ones

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 21 • BACKWARD 2 • Q3/5

642

QUESTION ID

S21-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

**D. 2 hundreds 9 tens
10 ones**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 21 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$400 - 158$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 21 • BACKWARD 2 • Q4/5

643

QUESTION ID

S21-B2-Q4

TYPE

Short answer

CORRECT ANSWER

242

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 21 • BACKWARD 2 • Q4/5

True or false?

6 tens 3 ones = 5 tens 3 ones

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 21 • BACKWARD 2 • Q5/5

644

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 4.NBT.B.4

FORWARD 1

Subtract decimals to hundredths • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$42.5 - 18.76$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • FORWARD 1 • Q1/5

646

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

23.74

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$9 - 3.45$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • FORWARD 1 • Q2/5

647

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • FORWARD 1 • Q2/5

Subtract.

$$6.03 - 2.8$$

- A.** 3.22
- B.** 3.75
- C.** 4.23
- D.** 3.23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • FORWARD 1 • Q3/5

648

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3.23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • FORWARD 1 • Q3/5

True or false?

$$10 - 0.25 = 9.75$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • FORWARD 1 • Q4/5

649

QUESTION ID

S21-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • FORWARD 1 • Q4/5

True or false?

$$5.4 - 2.15 = 3.35$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • FORWARD 1 • Q5/5

650

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 4.NBT.B.4

FORWARD 2

Subtract multi-digit decimals with the standard algorithm • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$100 - 36.875$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 2 • Q1/5

652

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

63.125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 2 • Q1/5

652

Subtract.

$$15.06 - 8.3$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 2 • Q2/5

653

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6.76

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 2 • Q2/5

Subtract.

$$4.2 - 1.375$$

- A.** 2.825
- B.** 2.925
- C.** 3.825
- D.** 3.175

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 2 • Q3/5

654

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2.825

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 2 • Q3/5

True or false?

$$50.5 - 0.505 = 49.995$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 2 • Q4/5

655

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 2 • Q4/5

True or false?

$$8 - 0.08 = 0.72$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 2 • Q5/5

656

QUESTION ID

S21-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 2 • Q5/5

Multiply a whole number of up to four digits by a one-digit number

MAIN	Grade 4 main question	4.NBT.B.5
BACKWARD 1	Multiplication facts within 100	3.OA.C.7
BACKWARD 2	Multiply one-digit numbers by multiples of 10	3.NBT.A.3
FORWARD 1	Multiply multi-digit whole numbers using the standard algorithm	5.NBT.B.5
FORWARD 2	Multiply multi-digit decimals	6.NS.B.3

SET 22 • 4.NBT.B.5

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$4,236 \times 7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 22 • MAIN • Q1/5

659

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

29,652

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 22 • MAIN • Q1/5

659

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$608 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 22 • MAIN • Q2/5

660

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

3,040

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 22 • MAIN • Q2/5

660

Multiply.

$$3,049 \times 6$$

- A. 3,055
- B. 18,254
- C. 18,924
- D. 18,294

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 22 • MAIN • Q3/5

661

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 18,294

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 22 • MAIN • Q3/5

A theater sells 1,275 tickets each night for 8 nights.

How many tickets does it sell?

Tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 22 • MAIN • Q4/5

662

QUESTION ID

S22-M-Q4

TYPE

Short answer

CORRECT ANSWER

10,200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 22 • MAIN • Q4/5

True or false?

$$2,513 \times 4 = 10,042$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 22 • MAIN • Q5/5

663

QUESTION ID

S22-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 22 • MAIN • Q5/5

SET 22 • 4.NBT.B.5

BACKWARD 1

Multiplication facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$7 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 1 • Q1/5

665

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 1 • Q1/5

True or false?

$$9 \times 8 = 72$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 1 • Q2/5

666

QUESTION ID

S22-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 1 • Q2/5

$$4 \times 7 = ?$$

- A.** 24
- B.** 32
- C.** 11
- D.** 28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 1 • Q3/5

667

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 1 • Q3/5

Multiply.

$$6 \times 9$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 1 • Q4/5

668

QUESTION ID

S22-B1-Q4

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 1 • Q4/5

True or false?

$$8 \times 7 = 54$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 1 • Q5/5

669

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 4.NBT.B.5

BACKWARD 2

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$7 \times 80$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 22 • BACKWARD 2 • Q1/5

671

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

560

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 22 • BACKWARD 2 • Q1/5

True or false?

$$5 \times 60 = 300$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 22 • BACKWARD 2 • Q2/5

672

QUESTION ID

S22-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 22 • BACKWARD 2 • Q2/5

$$3 \times 90 = ?$$

- A.** 27
- B.** 2,700
- C.** 93
- D.** 270

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 22 • BACKWARD 2 • Q3/5

673

QUESTION ID

S22-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 270

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 22 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$4 \times 70$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 22 • BACKWARD 2 • Q4/5

674

QUESTION ID

S22-B2-Q4

TYPE

Short answer

CORRECT ANSWER

280

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 22 • BACKWARD 2 • Q4/5

True or false?

$$8 \times 40 = 3,200$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 22 • BACKWARD 2 • Q5/5

675

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 4.NBT.B.5

FORWARD 1

Multiply multi-digit whole numbers using the standard algorithm • Grade 5 •
Number & Operations in Base Ten • 5.NBT.B.5

Use the standard algorithm to multiply. Show your work.

$$3,214 \times 26$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 22 • FORWARD 1 • Q1/5

677

QUESTION ID

S22-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

83,564

GRADING NOTE REQUIRED WORK: The standard algorithm: one partial product for each digit of the second factor (with regrouping), each lined up by place value, then the partial products added. Another method earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 22 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$507 \times 48$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 22 • FORWARD 1 • Q2/5

678

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

24,336

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 22 • FORWARD 1 • Q2/5

678

Multiply.

$$1,250 \times 34$$

- A. 42,500
- B. 8,750
- C. 42,050
- D. 41,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 22 • FORWARD 1 • Q3/5

679

QUESTION ID

S22-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 42,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 22 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$289 \times 75$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 22 • FORWARD 1 • Q4/5

680

QUESTION ID

S22-F1-Q4

TYPE

Short answer

CORRECT ANSWER

21,675

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 22 • FORWARD 1 • Q4/5

680

True or false?

$$406 \times 52 = 21,112$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 22 • FORWARD 1 • Q5/5

681

QUESTION ID

S22-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 4.NBT.B.5

FORWARD 2

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$4.236 \times 7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 2 • Q1/5

683

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

29.652

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$6.08 \times 0.5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 2 • Q2/5

684

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 2 • Q2/5

Multiply.

$$30.49 \times 0.6$$

- A. 18.94
- B. 18.294
- C. 182.94
- D. 1.8294

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 2 • Q3/5

685

QUESTION ID

S22-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 18.294

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 2 • Q3/5

True or false?

$$2.5 \times 1.6 = 4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 2 • Q4/5

686

QUESTION ID

S22-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 2 • Q4/5

True or false?

$$0.7 \times 0.08 = 0.56$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 2 • Q5/5

687

QUESTION ID

S22-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 2 • Q5/5

Multiply two two-digit numbers

MAIN

Grade 4 main question

4.NBT.B.5

BACKWARD 1

Multiply one-digit numbers by multiples of 10

3.NBT.A.3

BACKWARD 2

Break apart a factor to multiply

3.OA.B.5

FORWARD 1

Multiply multi-digit whole numbers using the standard algorithm

5.NBT.B.5

FORWARD 2

Multiply multi-digit decimals

6.NS.B.3

SET 23 • 4.NBT.B.5

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$34 \times 27$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 23 • MAIN • Q1/5

690

QUESTION ID

S23-M-Q1

TYPE

Short answer

CORRECT ANSWER

918

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 23 • MAIN • Q1/5

690

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$58 \times 63$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 23 • MAIN • Q2/5

691

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

3,654

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 23 • MAIN • Q2/5

691

Multiply.

$$46 \times 19$$

- A.** 414
- B.** 884
- C.** 864
- D.** 874

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 23 • MAIN • Q3/5

692

QUESTION ID

S23-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 874

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 23 • MAIN • Q3/5

A box holds 24 cans.

How many cans are in 36 boxes?

Cans: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 23 • MAIN • Q4/5

693

QUESTION ID

S23-M-Q4

TYPE

Short answer

CORRECT ANSWER

864

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 23 • MAIN • Q4/5

True or false?

$$72 \times 15 = 1,008$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 23 • MAIN • Q5/5

694

QUESTION ID

S23-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 23 • MAIN • Q5/5

SET 23 • 4.NBT.B.5

BACKWARD 1

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations
in Base Ten • 3.NBT.A.3

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3 \times 40$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 23 • BACKWARD 1 • Q1/5

696

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

120

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 23 • BACKWARD 1 • Q1/5

696

True or false?

$$6 \times 90 = 540$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 23 • BACKWARD 1 • Q2/5

697

QUESTION ID

S23-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 23 • BACKWARD 1 • Q2/5

$$8 \times 20 = ?$$

- A.** 28
- B.** 1,600
- C.** 160
- D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 23 • BACKWARD 1 • Q3/5

698

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 160

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 23 • BACKWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$9 \times 70$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 23 • BACKWARD 1 • Q4/5

699

QUESTION ID

S23-B1-Q4

TYPE

Short answer

CORRECT ANSWER

630

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 23 • BACKWARD 1 • Q4/5

True or false?

$$4 \times 50 = 2,000$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 23 • BACKWARD 1 • Q5/5

700

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 4.NBT.B.5

BACKWARD 2

Break apart a factor to multiply • Grade 3 • Operations & Algebraic Thinking •
3.OA.B.5

True or false?

$$8 \times 7 = 8 \times 5 + 8 \times 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

702

QUESTION ID

S23-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

$$6 \times 9 = 6 \times 5 + 6 \times \underline{\quad}$$

Blank: _____

ID S23-B2-Q2

703

S23-B2-Q2

Short answer

4

ID S23-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q2/5

Which expression is equal to 4×8 ?

- A. $4 \times 5 \times 3$
- B. $4 \times 5 + 4 \times 3$
- C. $4 \times 5 + 3$
- D. $4 + 5 \times 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

704

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4 \times 5 + 4 \times 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

Find 7×6 by adding $7 \times 3 + 7 \times 3$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q4/5

705

QUESTION ID

S23-B2-Q4

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q4/5

True or false?

$$9 \times 6 = 9 \times 3 + 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q5/5

706

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 4.NBT.B.5

FORWARD 1

Multiply multi-digit whole numbers using the standard algorithm • Grade 5 •
Number & Operations in Base Ten • 5.NBT.B.5

Use the standard algorithm to multiply. Show your work.

$$427 \times 53$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 23 • FORWARD 1 • Q1/5

708

QUESTION ID

S23-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

22,631

GRADING NOTE REQUIRED WORK: The standard algorithm: one partial product for each digit of the second factor (with regrouping), each lined up by place value, then the partial products added. Another method earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 23 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$86 \times 145$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 23 • FORWARD 1 • Q2/5

709

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12,470

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 23 • FORWARD 1 • Q2/5

Multiply.

$$312 \times 64$$

- A. 18,968
- B. 19,968
- C. 19,868
- D. 1,248

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 23 • FORWARD 1 • Q3/5

710

QUESTION ID

S23-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 19,968

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 23 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$2,005 \times 18$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 23 • FORWARD 1 • Q4/5

711

QUESTION ID

S23-F1-Q4

TYPE

Short answer

CORRECT ANSWER

36,090

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 23 • FORWARD 1 • Q4/5

711

True or false?

$$95 \times 95 = 9,025$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 23 • FORWARD 1 • Q5/5

712

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 23 • FORWARD 1 • Q5/5

SET 23 • 4.NBT.B.5

FORWARD 2

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3.4 \times 2.7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 23 • FORWARD 2 • Q1/5

714

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

9.18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 23 • FORWARD 2 • Q1/5

714

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.58 \times 6.3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 23 • FORWARD 2 • Q2/5

715

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3.654

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 23 • FORWARD 2 • Q2/5

715

Multiply.

$$4.6 \times 0.19$$

- A.** 4.79
- B.** 0.874
- C.** 0.0874
- D.** 8.74

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 23 • FORWARD 2 • Q3/5

716

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.874

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 23 • FORWARD 2 • Q3/5

True or false?

$$2.4 \times 3.6 = 8.64$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 23 • FORWARD 2 • Q4/5

717

QUESTION ID

S23-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 23 • FORWARD 2 • Q4/5

True or false?

$$7.2 \times 1.5 = 1.08$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 23 • FORWARD 2 • Q5/5

718

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 23 • FORWARD 2 • Q5/5

Illustrate and explain multiplication with equations, arrays, and area models

MAIN	Grade 4 main question	4.NBT.B.5
BACKWARD 1	Use area to show the distributive property	3.MD.C.7.c
BACKWARD 2	Break apart an array to multiply	3.OA.B.5
FORWARD 1	Use place value and area models to multiply decimals	5.NBT.B.7
FORWARD 2	Use the distributive property to write a sum as a product	6.NS.B.4

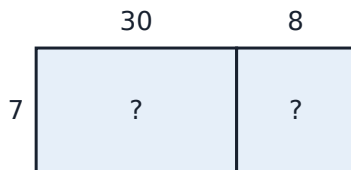
SET 24 • 4.NBT.B.5

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Complete the area model for 7×38 by writing the area of each part.

Then write the product.



Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 24 • MAIN • Q1/5

721

QUESTION ID

S24-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

**Drawing: Areas
written in the parts:
210 (7×30) and 56
(7×8)**

Written: 266

GRADING NOTE Grade both: the two areas must be 210 and 56, and the product $210 + 56 = 266$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

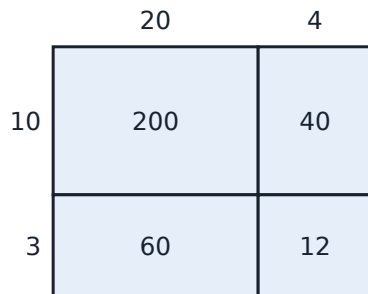
Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 24 • MAIN • Q1/5

The area model shows
 24×13 .

Write the four partial
products and the total.

Partial products: _____

Total: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 24 • MAIN • Q2/5

722

QUESTION ID

S24-M-Q2

TYPE

Short answer

CORRECT ANSWER

200, 40, 60, 12; 312

GRADING NOTE The partial products may be listed in any order. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 24 • MAIN • Q2/5

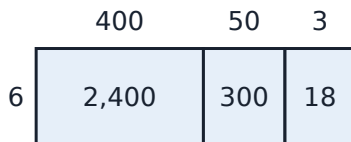
Which equation matches the area model?

A. $6 \times 453 = 2,400 + 50 + 3$

B. $6 \times 453 = 2,400 + 300 + 18$

C. $6 \times 453 = 6 \times 400 + 53$

D. $6 + 453 = 2,718$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 24 • MAIN • Q3/5

723

QUESTION ID

S24-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $6 \times 453 = 2,400 + 300 + 18$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 24 • MAIN • Q3/5

Use an area model or equations to multiply. Show your model or equations.

$$9 \times 2,146$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 24 • MAIN • Q4/5

724

QUESTION ID

S24-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

19,314

GRADING NOTE REQUIRED WORK: An area model or equations that break 2,146 into place-value parts, for example $9 \times 2,000 = 18,000$, $9 \times 100 = 900$, $9 \times 40 = 360$ and $9 \times 6 = 54$, then $18,000 + 900 + 360 + 54 = 19,314$. The answer without a valid model or equations earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 24 • MAIN • Q4/5

Explain why $15 \times 12 = 15 \times 10 + 15 \times 2$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 24 • MAIN • Q5/5

725

QUESTION ID

S24-M-Q5

TYPE

Short answer

CORRECT ANSWER

**12 = 10 + 2, so 15 groups of 12 are 15 groups of 10 plus 15 groups of 2;
150 + 30 = 180.**

GRADING NOTE Must show 12 split into $10 + 2$ and that multiplying each part and adding gives the same product. An array or area model split into 10 and 2 is also a correct explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 24 • MAIN • Q5/5

SET 24 • 4.NBT.B.5

BACKWARD 1

Use area to show the distributive property • Grade 3 • Measurement & Data •
3.MD.C.7.c

True or false?

A rectangle that is 4 by $(5 + 3)$ has an area of $4 \times 5 + 4 \times 3$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 24 • BACKWARD 1 • Q1/5

727

QUESTION ID

S24-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 24 • BACKWARD 1 • Q1/5

A 6-by-9 rectangle is split into a 6-by-5 part and a 6-by-4 part.

What is the total area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 24 • BACKWARD 1 • Q2/5

728

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

54 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 24 • BACKWARD 1 • Q2/5

A 3-by-8 rectangle is split into a 3-by-5 part and a 3-by-3 part.

Which expression gives its area?

A. $3 \times 8 + 3$

B. $3 \times 5 \times 3$

C. $3 + 5 + 3$

D. $3 \times 5 + 3 \times 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 24 • BACKWARD 1 • Q3/5

729

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $3 \times 5 + 3 \times 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 24 • BACKWARD 1 • Q3/5

$$7 \times 9 = 7 \times 5 + 7 \times \underline{\quad}$$

Blank: _____

ID S24-B1-Q4

730

S24-B1-Q4

Short answer

4

ID S24-B1-Q4

730

True or false?

$$5 \times 7 = 5 \times 4 + 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 24 • BACKWARD 1 • Q5/5

731

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 24 • BACKWARD 1 • Q5/5

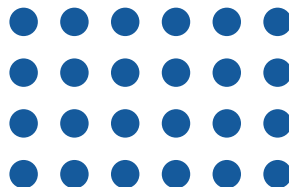
SET 24 • 4.NBT.B.5

BACKWARD 2

Break apart an array to multiply • Grade 3 • Operations & Algebraic Thinking •
3.OA.B.5

Write a multiplication equation for the array.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 24 • BACKWARD 2 • Q1/5

733

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4 × 6 = 24

GRADING NOTE $6 \times 4 = 24$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

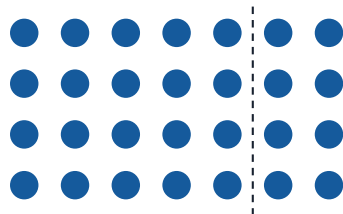
ID S24-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 24 • BACKWARD 2 • Q1/5

The dashed line splits the array into two parts.

Which equation matches?

- A. $4 \times 7 = 4 + 5 + 2$
- B. $4 \times 7 = 4 \times 5 \times 2$
- C. $4 \times 7 = 4 \times 5 + 4 \times 2$
- D. $4 + 7 = 4 \times 5 + 2$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 24 • BACKWARD 2 • Q2/5

734

QUESTION ID

S24-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C.

$$4 \times 7 = 4 \times 5 + 4 \times 2$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 24 • BACKWARD 2 • Q2/5

True or false?

An array of 3 rows of 9 can be split into 3 rows of 5 and 3 rows of 4.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 24 • BACKWARD 2 • Q3/5

735

QUESTION ID

S24-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 24 • BACKWARD 2 • Q3/5

An array of 6 rows of 8 is split into 6 rows of 5 and 6 rows of 3.

How many dots are in the whole array?

Dots: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 24 • BACKWARD 2 • Q4/5

736

QUESTION ID

S24-B2-Q4

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 24 • BACKWARD 2 • Q4/5

True or false?

An array of 5 rows of 6, split into two arrays of 5 rows of 3, has 33 dots.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 24 • BACKWARD 2 • Q5/5

737

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 4.NBT.B.5

FORWARD 1

Use place value and area models to multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

An area model splits 3×2.4 into 3×2 and 3×0.4 .

What is the product?

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 1 • Q1/5

739

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 1 • Q1/5

True or false?

$$4 \times 1.25 = 4 \times 1 + 4 \times 0.25 = 5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 1 • Q2/5

740

QUESTION ID

S24-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 1 • Q2/5

Which shows 6×3.5 broken apart by place value?

- A. $6 \times 3 \times 0.5$
- B. $6 \times 3 + 0.5$
- C. $6 + 3 + 0.5$
- D. $6 \times 3 + 6 \times 0.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 1 • Q3/5

741

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $6 \times 3 + 6 \times 0.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$5 \times 0.36$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 1 • Q4/5

742

QUESTION ID

S24-F1-Q4

TYPE

Short answer

CORRECT ANSWER

1.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 1 • Q4/5

742

True or false?

$$2 \times 4.3 = 8.3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 1 • Q5/5

743

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 4.NBT.B.5

FORWARD 2

Use the distributive property to write a sum as a product • Grade 6 • The Number System • 6.NS.B.4

Use the factor 4 to write $36 + 8$ as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 24 • FORWARD 2 • Q1/5

745

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4(9 + 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 24 • FORWARD 2 • Q1/5

True or false?

$$7(10 + 3) = 70 + 21$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 24 • FORWARD 2 • Q2/5

746

QUESTION ID

S24-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 24 • FORWARD 2 • Q2/5

Which expression equals 5×47 ?

- A. $5(40 + 7)$
- B. $5 \times 40 + 7$
- C. $5 + 40 \times 7$
- D. $(5 + 40) \times 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 24 • FORWARD 2 • Q3/5

747

QUESTION ID

S24-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $5(40 + 7)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 24 • FORWARD 2 • Q3/5

Use the greatest common factor to write $24 + 60$ as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 24 • FORWARD 2 • Q4/5

748

QUESTION ID

S24-F2-Q4

TYPE

Short answer

CORRECT ANSWER

12(2 + 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 24 • FORWARD 2 • Q4/5

True or false?

$$3(12 + 5) = 36 + 5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 24 • FORWARD 2 • Q5/5

749

QUESTION ID

S24-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 24 • FORWARD 2 • Q5/5

Divide up to four-digit dividends by one-digit divisors, with remainders

MAIN	Grade 4 main question	4.NBT.B.6
BACKWARD 1	Division facts within 100	3.OA.C.7
BACKWARD 2	Share equally	3.OA.A.2
FORWARD 1	Divide with two-digit divisors	5.NBT.B.6
FORWARD 2	Divide multi-digit numbers with the standard algorithm	6.NS.B.2

SET 25 • 4.NBT.B.6

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$952 \div 7$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 25 • MAIN • Q1/5

752

QUESTION ID

S25-M-Q1

TYPE

Short answer

CORRECT ANSWER

136

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 25 • MAIN • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$4,815 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 25 • MAIN • Q2/5

753

QUESTION ID

S25-M-Q2

TYPE

Short answer

CORRECT ANSWER

963

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 25 • MAIN • Q2/5

753

Divide.

$$2,184 \div 6$$

- A. 364
- B. 354
- C. 2,178
- D. 374

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 25 • MAIN • Q3/5

754

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 364

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 25 • MAIN • Q3/5

Divide. Write the quotient and the remainder.

$$1,357 \div 4$$

Quotient: _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 25 • MAIN • Q4/5

755

QUESTION ID

S25-M-Q4

TYPE

Short answer

CORRECT ANSWER

339 R 1

GRADING NOTE Quotient 339 and remainder 1. The question asks for a remainder, so a mixed number or decimal is not accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 25 • MAIN • Q4/5

True or false?

$$3,600 \div 9 = 40$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 25 • MAIN • Q5/5

756

QUESTION ID

S25-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 25 • MAIN • Q5/5

SET 25 • 4.NBT.B.6

BACKWARD 1

Division facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Divide.

$$42 \div 7$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 25 • BACKWARD 1 • Q1/5

758

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 25 • BACKWARD 1 • Q1/5

True or false?

$$72 \div 8 = 9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 25 • BACKWARD 1 • Q2/5

759

QUESTION ID

S25-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 25 • BACKWARD 1 • Q2/5

$$54 \div 6 = ?$$

- A.** 48
- B.** 7
- C.** 8
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 25 • BACKWARD 1 • Q3/5

760

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 25 • BACKWARD 1 • Q3/5

Divide.

$$27 \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 25 • BACKWARD 1 • Q4/5

761

QUESTION ID

S25-B1-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 25 • BACKWARD 1 • Q4/5

True or false?

$$63 \div 9 = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 25 • BACKWARD 1 • Q5/5

762

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 4.NBT.B.6

BACKWARD 2

Share equally • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

28 marbles are shared equally by 4 children.

How many marbles does each child get?

Marbles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 25 • BACKWARD 2 • Q1/5

764

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 25 • BACKWARD 2 • Q1/5

True or false?

When 45 is split into 5 equal groups, each group has 9.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 25 • BACKWARD 2 • Q2/5

765

QUESTION ID

S25-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 25 • BACKWARD 2 • Q2/5

Which expression shows 32 shared equally among 8 people?

A. $32 - 8$

B. 32×8

C. $32 \div 8$

D. $8 \div 32$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 25 • BACKWARD 2 • Q3/5

766

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $32 \div 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 25 • BACKWARD 2 • Q3/5

48 stickers are put into groups of 6.

How many groups are there?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 25 • BACKWARD 2 • Q4/5

767

QUESTION ID

S25-B2-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 25 • BACKWARD 2 • Q4/5

True or false?

20 shared equally by 4 people gives each person 16.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 25 • BACKWARD 2 • Q5/5

768

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 4.NBT.B.6

FORWARD 1

Divide with two-digit divisors • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$1,736 \div 28$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 25 • FORWARD 1 • Q1/5

770

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 25 • FORWARD 1 • Q1/5

770

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$5,904 \div 72$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 25 • FORWARD 1 • Q2/5

771

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

82

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 25 • FORWARD 1 • Q2/5

771

Divide.

$$3,360 \div 48$$

- A.** 70
- B.** 7
- C.** 68
- D.** 700

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 25 • FORWARD 1 • Q3/5

772

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 25 • FORWARD 1 • Q3/5

Divide. Write the quotient and the remainder.

$$2,000 \div 37$$

Quotient: _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 25 • FORWARD 1 • Q4/5

773

QUESTION ID

S25-F1-Q4

TYPE

Short answer

CORRECT ANSWER

54 R 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 25 • FORWARD 1 • Q4/5

True or false?

$$1,125 \div 25 = 54$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 25 • FORWARD 1 • Q5/5

774

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 25 • FORWARD 1 • Q5/5

SET 25 • 4.NBT.B.6

FORWARD 2

Divide multi-digit numbers with the standard algorithm • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$9,516 \div 39$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 25 • FORWARD 2 • Q1/5

776

QUESTION ID

S25-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

244

GRADING NOTE REQUIRED WORK: The standard division algorithm with every step shown (divide, multiply, subtract, bring down). A correct quotient without the algorithm earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 25 • FORWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$27,648 \div 72$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 25 • FORWARD 2 • Q2/5

777

QUESTION ID

S25-F2-Q2

TYPE

Short answer

CORRECT ANSWER

384

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 25 • FORWARD 2 • Q2/5

777

Divide.

$$15,525 \div 45$$

- A. 355
- B. 345
- C. 3,450
- D. 335

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 25 • FORWARD 2 • Q3/5

778

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 345

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 25 • FORWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$8,184 \div 62$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 25 • FORWARD 2 • Q4/5

779

QUESTION ID

S25-F2-Q4

TYPE

Short answer

CORRECT ANSWER

132

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 25 • FORWARD 2 • Q4/5

779

True or false?

$$20,808 \div 51 = 408$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 25 • FORWARD 2 • Q5/5

780

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 25 • FORWARD 2 • Q5/5

Illustrate and explain division with equations, area models, and place value

MAIN	Grade 4 main question	4.NBT.B.6
BACKWARD 1	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 2	Find a missing side from the area of a rectangle	3.MD.C.7.b
FORWARD 1	Explain division with two-digit divisors using area models and equ...	5.NBT.B.6
FORWARD 2	Connect the standard division algorithm to partial quotients	6.NS.B.2

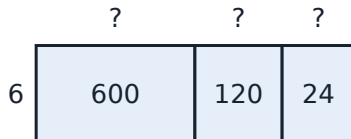
SET 26 • 4.NBT.B.6

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Complete the area model for $744 \div 6$ by writing the missing length of each part.

Then write the quotient.



Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 26 • MAIN • Q1/5

783

QUESTION ID

S26-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

**Drawing: Lengths
written on the parts:
100, 20 and 4**

Written: 124

GRADING NOTE Grade both: the lengths 100, 20 and 4 ($6 \times 100 = 600$, $6 \times 20 = 120$, $6 \times 4 = 24$), and the quotient 124.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 26 • MAIN • Q1/5

Explain how these equations show $852 \div 4$.

$$4 \times 200 = 800$$

$$4 \times 13 = 52$$

$$800 + 52 = 852$$

Quotient: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 26 • MAIN • Q2/5

784

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

213

GRADING NOTE Explanation: 852 is split into 800 and 52; 4 goes into them 200 times and 13 times, so 4 goes into 852 a total of 213 times.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 26 • MAIN • Q2/5

Ana finds $3,515 \div 5$ by thinking $3,500 \div 5 = 700$ and $15 \div 5 = 3$.

What is the quotient?

- A.** 7,003 **B.** 73
C. 703 **D.** 730

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 26 • MAIN • Q3/5

785

QUESTION ID

S26-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 703

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 26 • MAIN • Q3/5

Use an area model or equations to divide. Show your model or equations.

$$1,792 \div 8$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 26 • MAIN • Q4/5

786

QUESTION ID

S26-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

224

GRADING NOTE REQUIRED WORK: An area model or equations that split 1,792 into parts that 8 divides, for example $8 \times 200 = 1,600$ and $8 \times 24 = 192$, with $1,600 + 192 = 1,792$, so the quotient is 224. The answer without a valid model or equations earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 26 • MAIN • Q4/5

True or false?

To find $636 \div 3$, you can divide each place:
 $600 \div 3 = 200$, $30 \div 3 = 10$, and $6 \div 3 = 2$. The quotient is 212.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 26 • MAIN • Q5/5

787

QUESTION ID

S26-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 26 • MAIN • Q5/5

SET 26 • 4.NBT.B.6

BACKWARD 1

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $56 \div 8$ by thinking $8 \times ? = 56$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 26 • BACKWARD 1 • Q1/5

789

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 26 • BACKWARD 1 • Q1/5

True or false?

$36 \div 9 = 4$ because $9 \times 4 = 36$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 26 • BACKWARD 1 • Q2/5

790

QUESTION ID

S26-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 26 • BACKWARD 1 • Q2/5

Which fact helps you find $45 \div 5$?

- A.** $5 \times 9 = 45$
- B.** $45 - 5 = 40$
- C.** $9 + 5 = 14$
- D.** $5 \times 5 = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 26 • BACKWARD 1 • Q3/5

791

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $5 \times 9 = 45$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 26 • BACKWARD 1 • Q3/5

Find the unknown factor.

$$7 \times ? = 63$$

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 26 • BACKWARD 1 • Q4/5

792

QUESTION ID

S26-B1-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 26 • BACKWARD 1 • Q4/5

True or false?

$24 \div 3 = 6$ because $3 \times 6 = 24$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 26 • BACKWARD 1 • Q5/5

793

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 4.NBT.B.6

BACKWARD 2

Find a missing side from the area of a rectangle • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle has an area of 48 square units. One side is 6 units long.

How long is the other side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 26 • BACKWARD 2 • Q1/5

795

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 26 • BACKWARD 2 • Q1/5

True or false?

A rectangle that is 7 units by 5 units has an area of 35 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 26 • BACKWARD 2 • Q2/5

796

QUESTION ID

S26-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 26 • BACKWARD 2 • Q2/5

A rectangle has an area of 36 square feet and a width of 4 feet.

What is its length?

- A.** 32 feet **B.** 40 feet
C. 9 feet **D.** 8 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 26 • BACKWARD 2 • Q3/5

797

QUESTION ID

S26-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 9 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 26 • BACKWARD 2 • Q3/5

What is the area of a rectangle that is 8 cm by 6 cm?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 26 • BACKWARD 2 • Q4/5

798

QUESTION ID

S26-B2-Q4

TYPE

Short answer

CORRECT ANSWER

48 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 26 • BACKWARD 2 • Q4/5

True or false?

A rectangle with an area of 30 square units and one side of 5 units has another side of 25 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 26 • BACKWARD 2 • Q5/5

799

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 4.NBT.B.6

FORWARD 1

Explain division with two-digit divisors using area models and equations •
Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6